Selective Immigration Policies and the U.S. Labor Market

Joan Llull*§
Institute of Economic Analysis (IAE-CSIC)
and Barcelona School of Economics

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While immigration of unskilled workers often generates controversy in the political arena, there is often more consensus in favor of selective immigration policies. This paper studies the effects of selective immigration policies on the labor market. High skilled immigration introduces two potentially confronting forces on labor market prospects of native workers: first, it increases the competition for skilled jobs, reducing labor market opportunities, and, as a result, reducing native incentives to invest in human capital; second, it increases productivity through spillovers and technological progress. I pose and estimate a labor market equilibrium dynamic discrete choice model that can account for these effects. The estimated model is used to evaluate the labor market consequences of the two most important skill-biased immigration policies in recent U.S. history: the introduction of H-1B visa program in 1990, and the elimination of the National Origins Formula in 1965. I also use the model to simulate the level of selectivity of immigration policy that maximizes native workers' welfare.

I. Introduction

Immigration policy has been at the core of the political debate in many developed countries over the last decades. Historically, a large part of this debate concerned, mainly, the immigration of low skilled workers, and the inflow of refugees, but high-skill immigration policies were subject to less controversy, and gathered more support. Immigration policies in Canada, Australia, New Zealand, Japan,

* IAE-CSIC. Campus de Bellaterra, 08193, Bellaterra, Cerdanyola del Vallès, Barcelona (Spain). URL: https://joanllull.github.io. E-mail: joan.llull [at] iae.csic [dot] es.

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and the United Kingdom are mostly based on points systems that favor immigration of skilled workers, and Germany recently implemented a similar policy. In the United States, the introduction of the H1-B visa in 1990 drastically increased the inflow of high skilled workers, systematically reaching the annual maximum quota every year. In his first mandate, President Trump questioned the efficacy of the H-1B program in bringing in high quality immigration, and announced changes to prevent fraud and to make it more selective, but did not threaten its existence.[1]

The Syrian refugee crisis has put immigration policy back at the core of the political debate in many developed countries. The possibility of exerting a larger control on immigration and of having stronger borders was one of the main arguments used to support Brexit, it was a salient issue in President Donald J. Trump's presidential campaign, who proposed the construction of a wall on the Mexico-United States border, and it has been central in several (some of them successful) presidential campaigns in many European countries.[2] Unlike with general immigration, the policies that favor the inflow of highly skilled workers are generally less challenged. Immigration policies in Canada, Australia, and the United Kingdom are mostly based on points systems that favor immigration of skilled workers, and Japan recently implemented a similar policy. Canada recently launched as well the Global Skills Strategy, a program that gives Canadian employers faster access to highly skilled foreign workers. In the United States, the H-1B visa program and some of the executive actions approved by President Barack Obama in November 2014 aimed at facilitating immigration of highly skilled workers. And even

Are there economic gains of selective immigration policies relative to general immigration? Whose labor market prospects are improved by skilled immigration, whose are worsened, and by how much? Does high skilled immigration discourage native investments in human capital? Are there knowledge spillovers from skilled immigration? What level of selectivity in immigration policy maximizes natives' utility? The answers to these questions are fundamental for the design of immigration policy. In this paper, I provide answers to these questions by

[1] In April 2017, President Trump signed the executive order "Buy American and Hire American" which dictates, among other things, that "in order to promote the proper functioning of the H-1B visa program, the Secretary of State, the Attorney General, the Secretary of Labor, and the Secretary of Homeland Security shall, as soon as practicable, suggest reforms to help ensure that H-1B visas are awarded to the most-skilled or highest-paid petition beneficiaries". (The White House Press Office, April 18, 2017).

[2] For example, Jörg Haider and Norbert Hofer in Austria, Gábor Vona and Viktor Orban in Hungary, Geert Wilders in the Netherlands, Nikos Michaloliakos in Greece, Timo Soini in Finland, Frauke Petry in Germany, Kristian Thulesen Dahl in Denmark, Gianluca Iannone in Italy, Björn Söder in Sweden, and Marine Le Pen in France.

posing and estimating a labor market equilibrium dynamic discrete choice model. The estimated model is used to simulate a different set of positive and normative counterfactual exercises that provide the relevant answers. The positive analysis focuses on the first four questions. These simulations evaluate the economic impact of the two most important skill-biased immigration policies in recent U.S. history: the introduction of the H-1B visa program in 1990, and the elimination of the National Origins Formula in 1965.[3] In the normative analysis, I quantify the level of selectivity of the immigration policy that maximizes the wellbeing of native workers. On the one hand, attracting more skilled workers enhance native productivity through externalities and knowledge spillovers. On the other hand, skilled immigration poses competition on skilled natives, which may discourage investments in the first place.

My modeling approach builds on Llull (2018a), who shows the importance of accounting for native human capital and labor supply adjustments to quantify labor market effects of the increase in (mostly unskilled) immigration in the United States over the last four decades. Unlike Llull (2018a), the estimated model accounts for two confronting forces of high skilled immigration in the labor market. First, the model allows for a potential externality through knowledge spillovers and endogenous technological progress. The model features the generation of ideas as a consequence of using skilled labor (and equipment capital) in production. This generation of ideas and knowledge spillovers endogenously produce neutral and skill-biased technological change, which affects the productivity of other labor market inputs (and capital). The second one, is a competition effect through the labor market equilibrium. The inflow of skilled workers puts downward pressure on wages of competing workers (the closest substitutes are skilled natives), reducing their incentives to invest in human capital in the first place. These two confronting forces make the determination of the level of selectivity in the immigration policy that native workers are willing to accept an empirical question.

The estimated model is a labor market equilibrium model with knowledge spillovers. On the labor demand side, a representative firm (which represents

[3] The H-1B visa program is a guest program targeted to attract workers in Science, Technology, Engineering and Mathematics (STEM) fields for a once-renewable period of three years, after which the employer can sponsor the worker for permanent residency. The main users of this program are immigrants from India, China, and other Asian countries. The National Origins Formula was a quota system in place between 1924 and 1965 that selected immigrants on the basis of national origin in order to preserve the ethnic mix of the U.S. population. The removal of the National Origins Formula in 1965 completely reshaped the skill composition of immigrants in the U.S., switching from a relatively educated Western immigration to a less educated Latin American and then Asian one (Borjas, 1993, 1995; Antecol, Cobb-Clark and Trejo, 2003).

the behavior of a continuum of atomistic firms) combines three types of labor (blue collar, white collar, and STEM) and two types of capital (structures and equipment) to produce a single output. In doing so, the firm generates ideas as a by-product of using equipment capital and STEM labor in production. Ideas generate productivity spillovers changing the relative demand for different inputs, and also fostering factor neutral technological progress. As a representation of atomistic firms, the representative firm does not take into account these productivity enhancements in its labor demand, and, therefore, this constitutes an externality. On the labor supply side, heterogeneous individuals decide on education, participation, and occupation over their life cycle. Individuals are heterogeneous in many dimensions, including, in the case of immigrants, national origin, which is essential both to determine the prevalence of H-1B visas, and to simulate the National Origins Formula.

The model is estimated using U.S. micro-data from both the Current Population Survey (CPS) for years 1993–2023 (March Supplements, linked over two consecutive years) along with national-level data for some of the aggregate variables. One of the central aspects of this paper is to credibly identify the knowledge externality. Beyond functional form assumptions, and econometric implementation, it is fundamental to understand the variation from the data that identifies the spillovers. To this end, it is crucial to have a measurement of the stock of ideas. In this paper, I use two alternative variables to measure it: the accumulation of intellectual property products (IPP) capitalized in the National Income and Product Accounts (NIPA) recently revised by the Bureau of Economic Analysis (BEA), and the number of patents generated in the U.S. in a given year. Armed with such measurements, the model interprets the data as follows. Changes in wages that follow an exogenous change in STEM labor supply (or capital equipment) holding fixed the stock of ideas (e.g. a negative "ideas shock" offsets the change in labor inputs) are interpreted as elasticities of substitution across labor inputs. On the contrary, changes in the stock of ideas that do not follow any change in labor or capital inputs identify the externality.

The estimation of the model using full solution methods (like in Lee and Wolpin, 2006, 2010; Llull, 2018a, 2021) is computationally too demanding. Alternatively, I estimate the model using conditional choice probability (CCP) estimation, combining techniques and arguments developed by Hotz and Miller (1993), Hotz, Miller, Sanders and Smith (1994), Altuğ and Miller (1998), Aguirregabiria and Mira (2002), and Arcidiacono and Miller (2011) (see Arcidiacono and Ellickson (2011) for a review). CCP estimation avoids solving for the value functions in each iteration of the parameter search, which is very costly. Additionally, it has the advantage of transparently presenting parameter identification, as well as allowing for sensitivity analysis to different functional form assumptions and different datasets.

Altuğ and Miller (1998) is the first (and, to my knowledge, only) paper that applies CCP estimation methods to models that feature aggregate shocks. They present a labor supply (and consumption) decision model that allows for aggregate conditions that are consistent with Pareto optimal allocations. As these authors note, the implementation of this class of CCP estimators requires estimates for the CCPs for different (counterfactual) realizations of the aggregate shocks in order to correctly specify expectations about the future. Altuğ and Miller (1998) obtain these CCPs exploiting the variation in the shadow value of consumption, heterogeneous across individuals, along with stationarity. For example, one can predict the behavior of a wealthy individual living in an economic slump by observing the behavior of a poorer individual living in a prosperous world. They can use this argument in identification because they have data on consumption. Absent consumption information in my data, I show that one can still exploit the stationarity of the model along with the equilibrium structure to infer the counterfactual CCPs from time series variation in aggregate conditions. Given stationarity, calendar time only affects labor supply decisions through skill prices (driven by the aggregate shock), and thus constitutes a sufficient statistic for the aggregate shock in the observed baseline economy. This allows me to recover equilibrium skill prices by estimating the wage equations using current period baseline CCPs. Having recovered them, I reestimate the CCPs (now conditional on skill prices). In doing so, I interpret periods with high skill prices as counterfactuals for periods with low prices, had these prices been high.

This paper contributes to the growing literature on the consequences of skilled immigration.[4] Different strands of this literature analyze: the effect of skilled immigration on patenting and entrepreneurship (Hunt and Gauthier-Loiselle, 2010; Kerr and Lincoln, 2010; Hunt, 2011); the displacement effect on native STEM employment and wages (Borjas, 2009; Kerr and Kerr, 2013; Bound, Braga, Golden and Khanna, 2015; Peri, Shih and Sparber, 2015; Kerr, Kerr and Lincoln, 2015; Doran, Gelber and Isen, 2022; Bound, Khanna and Morales, 2018; Ma, 2020); the career prospects of science and technology immigrants relative to similar natives

[4] More generally, it also contributes to the general literature on immigration on wages (Borjas, 2003; Card, 2009; Ottaviano and Peri, 2012; Dustmann, Frattini and Preston, 2013; Llull, 2018a,c, 2021; see Dustmann, Schönberg and Stuhler, 2016 for a recent survey).

(Gaulé and Piacentini, 2013; Hunt, 2015); the crowding out effects of H-1B visas on native students (Kato and Sparber, 2013; Orrenius and Zavodny, 2015); and the competition and spillover effects in the space of ideas, exploiting evidence on massive inflows of foreign professors (Borjas and Doran, 2012, 2015a,b, and Moser, Voena and Waldinger, 2014).

Among these strands of the literature, my paper is mostly related to the first two.[5] Except for Bound et al. (2015), Bound et al. (2018), and Ma (2020), all these papers use a reduced form approach. The structural approach allows me to expand this literature in several dimensions. First, I take into account potential displacement effects not only on employment of natives, but also on their decisions to invest in education and to become STEM workers. Second, the structural model allows me to identify the spillover effects on wages through the generation of knowledge, measured as the accumulation of patents, or IPP capital. For example, while Hunt and Gauthier-Loiselle (2010) and Kerr and Lincoln (2010) analyze the effect of skilled immigration on patenting; I additionally measure how this extra intangible capital maps into higher wages for natives, and how that affects their incentives in the labor market. Third, I quantify the heterogeneous effects of size and selectivity of immigration policy across different groups of native workers and I use the estimated model to characterize the level of selectivity of the immigration policy that maximizes native workers' wellbeing. Four, I take into account capital-skill complementarity, which is important to correctly measure the wage impact of skilled (and unskilled) immigration (see Lewis, 2011, 2013, and Llull, 2018a, 2021), and I allow for skill-biased technological spillovers from high skilled immigration. And five, on top of the H-1B visa policy, the estimated model is also used to evaluate the National Origins Formula as a selective immigration policy.

Bound et al. (2015) present a macro-calibrated model of partial equilibrium for the market of computer scientists. Even though they focus on a much narrower market (computer scientists), their paper is more related to mine than the others above as it takes into account the change in the supply of prospective workers with computer science majors (education). It also allows, in a reduced form way, for spillovers of immigration on overall productivity, even though their mechanism is not endogenous. Bound, Khanna and Morales (2018) expand their previous model to endogenize technological change, linking productivity increases in the

[5] It is also related to the last strand, even though this group studies a narrower population of interest, namely Soviet mathematicians migrating to the U.S. after the collapse of the Soviet Union, and Jewish chemists migrating from Nazi Germany. The model is also able to speak to the remaining two strands, but they are less central to the main contribution of the paper.

U.S. during the 1990s to increase in the utilization of computer scientists in the economy. Ma (2020) goes beyond the computer scientist market, and estimates a model in which natives and immigrants can work in either computer science or in other STEM jobs. She studies the potential crowding out of natives into other STEM fields, which is the differentiation mechanism that allows them avoid competition from immigrants. She, however, abstracts from knowledge spillovers, and from effects on non-STEM occupations.

More broadly, my paper is also related to other literatures. First, it is related to the literature that estimates dynamic labor market equilibrium models of career choices (Altuğ and Miller, 1998; Heckman, Lochner and Taber, 1998; Lee, 2005; Lee and Wolpin, 2006, 2010; Llull, 2018a, 2021). Moreover, it contributes to the literatures that analyze skill-biased technical change and capital-skill complementarity (e.g., Krusell, Ohanian, Ríos-Rull and Violante, 2000; see Acemoglu, 2002; Acemoglu and Autor, 2011 for surveys), and knowledge spillovers in aggregate production (Romer, 1986; Lucas, 1988; see Klenow and Rodríguez-Clare, 2005 for a review), providing a model that features endogenous neutral and skill-biased technological change. Finally, it relates to the macro literature that explores the role of intangible capital in explaining the recent evolution of labor productivity and the labor share (e.g., McGrattan and Prescott, 2010, 2014; Koh, Santaeulàlia-Llopis and Zheng, 2016).

The rest of the paper is organized as follows. Section II introduces some policy background and descriptive statistics. Section III presents the model. Section IV discusses identification. Section V introduces the estimation procedure. Section VI presents the estimated parameters of the model and evaluates the goodness of fit. Section VII presents simulation results for the different policy experiments. And Section VIII concludes.

II. Immigration in the United States: Policy and Facts

A. U.S. Immigration Policy Background: From the National Origins Formula to the H-1B Visa Program

Throughout its history, the United States has been a nation of immigrants. From colonial times to mid-nineteenth century Western European immigrants (especially from Britain and Ireland, but also from Germany and Scandinavia) kept entering the U.S. without any federal legislation (Ewing, 2012). Beginning in 1850s, the so-called "new immigration" brought in immigrants from Eastern and Southern Europe as well as from Asia and Russia. Americans' preference for "old" rather than "new" immigration reflected a sudden rise in conservatism and the appearance of the first nativist movements. In 1875 the first federal immigration law was passed; this law prohibited the entrance of criminals and convicts, prostitutes, and Chinese contract laborers. This law paved the road for the 1882 Chinese Exclusion Act, which almost prohibited Chinese workers to enter the United States.[6] It was the first of many laws that targeted specific ethnic groups, starting a bias against Asian that lasted until 1952.[7]

The Immigration Act of 1917 defined a "barred zone" of nations in the Asia-Pacific triangle from which immigration was prohibited. In 1921 the U.S. Congress passed the Emergency Quota Act, which limited the annual number of immigrants to be admitted from any country to a maximum of the 3% of the number of persons from that country living in the U.S. in 1910; in 1924, the share was reduced to 2% and the reference year was switched to 1890. It was the birth of the National Origins Formula. The Immigration and Nationality Act of 1952 consolidated this system setting the quotas for each country to one sixth of one percent of the number of persons of that ancestry living in the United States as of 1920. These restrictions, aimed at preserving the ethnic composition of U.S. population, reserved most immigration slots for immigrants from the United Kingdom, Ireland, and Germany (Ewing, 2012).

The 1965 Amendments to the Immigration and Nationality Act radically changed U.S. immigration policy. The National Origins Formula was abolished, and replaced by aggregate limitations (initially by hemisphere, worldwide from 1976) with a maximum amount per country (common to all of them). The new policy also allowed to issue an unlimited amount of visas to immediate relatives (parents, spouses and children) of U.S. citizens and legal immigrants.[8]

The 1965 Amendments were not aimed at fostering immigration or changing the ethnic composition of immigrant inflows. They were rather a reaction to the civil rights movements during 1960s. According to the speech by President Lyndon B.

[6] Later on, Chinese were issued Japanese passports to enter the United States. In 1907, a "Gentleman's Agreement" with Japan effectively ended with Chinese and Japanese immigration.

[7] The Immigration and Naturalization Act of 1952 was the first step towards removing racial distinctions from U.S. immigration policies.

[8] The quotas from the National Origins Formula were not of application for the Western Hemisphere (Canada, Latin America, and the Caribbean). However, the legal immigration process from Latin America and the Caribbean was very costly. These large costs kept immigration from these countries not very far from the levels that would be implied the quotas. In 1942, the U.S. government introduced a large-scale program of temporary immigration of Mexican workers, the so-called *bracero* program. The costly process of immigrating permanently to the U.S. fostered an important increase in unauthorized immigration (through *braceros*' overstays). In 1954, under the "Operation Wetback", about one million Mexican immigrants were deported (Ewing, 2012). Even though this program ended in 1964, the introduction of the family reunification visa completely transformed the immigration process from Mexico.

Johnson when he signed the legislation into law, the reform was not "revolutionary":"it does not change the lives of millions" he said. Several of the promoters of the reform defended it in the debate at the U.S. Senate. Senator Edward M. Kennedy enumerated what, according to his view, the policy would not do: "First, our cities will not be flooded with a million of immigrants annually. [...] Secondly, the ethnic mix of this country will not be upset". Several representatives expressed the same opinion.[9] Many of them, emphasized that "the proposed legislation would not greatly increase the number of immigrants" (Senator Eugene McCarthy) and highlighted that the ethnic mix would not change: "the people from that part of the world [the Asia-Pacific Triangle] probably will never reach 1 percent of the U.S. population" (Senator Hiram Fong). However, as we discuss in Section II.B, removing the National Origins Formula not only changed the ethnic mix of the country drastically, but also precluded the radical change in the skill composition of immigrant inflows that followed.

After subsequent policies mostly focused on preventing illegal immigration (e.g., the 1986 Immigration Reform and Control Act (IRCA), followed by an amnesty, and the 1996 Illegal Immigration Reform and Immigrant Responsibility Act), one of the most important policy changes after 1965 came with the 1990 Immigration Act which restricted the number of visas to be issued to immediate relatives of previous immigrants and U.S. citizens, and established a preference skilled immigration. This policy introduced the H-1B visa for guest skilled immigrant workers in "specialty occupations". These occupations are defined as requiring theoretical and practical application of a body of highly specialized knowledge in a field of human endeavor, including, but not limited to, architecture, engineering, mathematics, physical sciences, social sciences, medicine and health, education, law, accounting, business specialties, theology, and art (Bound et al., 2015). Applicants have an educational requirement of at least a bachelor's degree. H-1B visas restrict holders to work only for the company that sponsored them. The standard duration of an H-1B visa is for a stay of three years, renewable up to a maximum of 6 years. However, firms can sponsor H-1B visa holders for a permanent resident visa.[10] Since 1990 there has been a cap in the total number of H-1B visas that

[9] Among others, interventions along these lines included those from Senators Edward and Robert Kennedy, Hart, Fong, Scott, Pell, Williams, Kuchel, Bartlett, Inouye, McCarthy, McNamara, Moss, Proxmire, the Secretary of Labor Willard Wirtz, and the Secretary of State Dean Rusk. Their interventions are transcribed in the Senate Part 1, Book 1 as made available at http://vdare.com/articles/so-much-for-promises-quotes-re-1965-immigration-act, accessed March 23th, 2017.

[10] Before H-1B, the Immigration and Nationality Act of 1952 introduced the H-1 visa, targeted at guest workers of "distinguished merit and ability" (Bound et al., 2015). However, they

can be issued, which has been binding since mid 1990s (except for years 2000–2003, in which the cap was threefold increased). Since year 2000, universities and non-profit research facilities are excluded from the cap. The H-1B visa has been instrumental in the recent increase in highly skilled immigration, especially from India, but also, to a lesser extent, from China and other parts of Asia.

B. Ethnic and Skill Composition of Immigration

This section provides descriptive statistics that offer a general picture about the evolution of the skill composition of immigration in the United States over the recent decades. It also shows the evolution of the national origin composition, demonstrating that the two are highly associated. Finally, it provides evidence of the increasing incidence of immigration in STEM occupations.

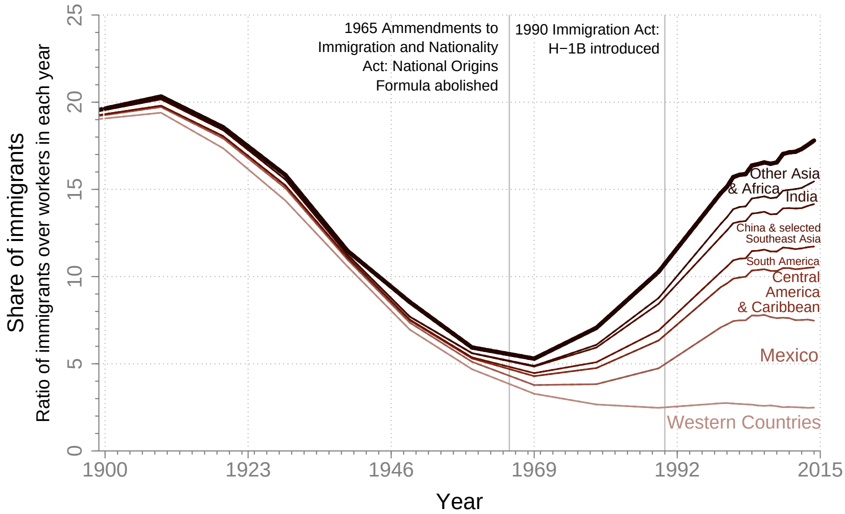
Figure 1 shows the evolution of the share of immigrants in the population working-age by national origin over the last century. The figure shows that the predictions of the promoters of the 1965 Amendments to the Immigration and Naturalization Act were not very accurate. From the time the legislation was enacted until the present day, the stock of immigrants aged 16 to 64 in the U.S. increased by a factor of six, roughly from 6 to 37 million immigrants. This average inflow of about 650,000 immigrants of per year increased the share of immigrants in the population working-age from around 5.5% to 17.8%. Furthermore, the ethnic mix changed substantially. By mid 1960s, the majority of working-age immigrants were from Western countries (70.9%), but the inflows from these countries, which by then had already been decreasing for decades, continued decreasing until today. By 2015, Western immigrants only represent 14% of all immigrants working-age. On the contrary, coinciding with the approval of the 1965 bill, a steady inflow of Mexican and other Central American/Caribbean immigrants increased their presence from 8.2% and 6.3% to 28% and 17.1% respectively. More recently, the inflow of Asian immigrants has increased substantially, in particular, from China and selected Southeast Asia, and, especially, India.[11] In 1990, when H-1B visa program was introduced, Indian immigrants were only 3.1% of all working-age immigrants, while in 2015 they represent 7.3% of the total.

The national origin composition of immigration is closely associated to the skills of immigrants. Table 1 explores the extent to which this is the case. Panel A re-

never became as popular as H-1B visas because the H-1B program introduced the possibility of transferring to permanent immigrant status, which made it more appealing for workers and firms.

[11] The selected set of Southeast Asian countries grouped together with China includes Taiwan, Hong Kong, South Korea, Japan, the Philippines, and Singapore. Hereinafter, I refer to this group as selected Southeast Asia.

FIGURE 1. IMMIGRANT SHARE BY NATIONAL ORIGIN (1900-2015)



Note: Areas delimited by plotted lines show the share that immigrants from each national origin represent of all individuals of working-age. Country grouping and inter-census interpolations are described in Appendix A. *Sources:* Census data (1900-2000) and ACS (2001-2015).

ports average years of education for natives, for immigrants as a whole, and for immigrants from each national origin. Panel B reports the fraction of individuals in each group that has a college degree. Both panels describe a similar picture. In 1970, natives and immigrants had similar education levels. However, since then, education of immigrants increased at a slower rate than that of natives. Interestingly, the change in the national origin composition of immigration that followed the removal of the National Origins Formula is crucially associated with this slower increase. As noted in Figure 1, a massive increase in immigration from Mexico and Central American and Caribbean countries followed the approval of the 1965 bill. These two groups of countries have, by far, the lowest education levels, which pushes the average education level of immigrants down. Once disaggregated by national origin, the average education level of immigrants from each of the origin country groups evolved with a similar slope than natives. Moreover, besides Mexican and Central American immigration, other groups also have education levels that are similar to (or even higher than) natives. Table 1 also shows evidence of the influence of the introduction of H-1B visas on the education level of immigrants. Both at the aggregate level and for the majority of country groups individually, there is an important increase in education (relative to natives) during the 1990s, coinciding with the introduction of these visas. On aggregate, immigrants increased average education by one year, while natives only did by 0.7 years, despite the aforementioned increase in the importance of Mexico and Central American countries over the period. At the country group level, average education increased, over that decade, by 1.2, 1.5, 0.9, 0.9, 0.7, 0.4, and

Table 1—Education of Natives and Immigrants

	1970	1980	1990	2000	2010	2015
A. Average years of education:						
Natives	11.2	12.3	13.0	13.7	13.7	13.8
Immigrants	11.0	11.7	12.1	13.1	13.1	13.3
Western countries	10.5	11.8	12.8	14.0	14.3	14.6
Mexico	6.3	7.1	7.6	9.1	9.3	9.6
Central America & Caribbean	10.2	10.9	11.0	11.9	11.5	11.5
South America	11.5	12.1	12.6	13.5	13.4	13.6
China & sel. Southeast Asia	11.7	13.2	13.7	14.4	14.6	14.6
India	15.6	15.2	15.1	15.5	15.4	15.4
Other Asia & Africa	11.2	11.8	12.2	13.3	13.2	13.5
B. Fraction with a college degree (%):						
Natives	23.2	35.3	50.8	57.7	56.1	58.2
Immigrants	23.6	35.5	44.3	47.1	45.3	47.7
Western countries	22.1	35.2	51.3	61.2	64.8	69.1
Mexico	6.6	10.1	13.8	16.1	15.4	17.0
Central America & Caribbean	22.0	29.7	35.0	37.8	34.2	35.0
South America	33.5	40.6	48.4	56.2	52.9	55.9
China & sel. Southeast Asia	42.9	57.3	65.9	71.1	71.9	72.1
India	80.7	78.7	77.4	79.1	80.5	79.3
Other Asia & Africa	32.5	41.4	51.4	59.5	58.1	60.5

Note: Figures in each panel indicate respectively the average years of education and the percentage of individuals with a college degree in each group. The sample is restricted to individuals aged 24-64. *Sources*: Census data (1970-2000) and ACS (2009-2011, and 2014-2015).

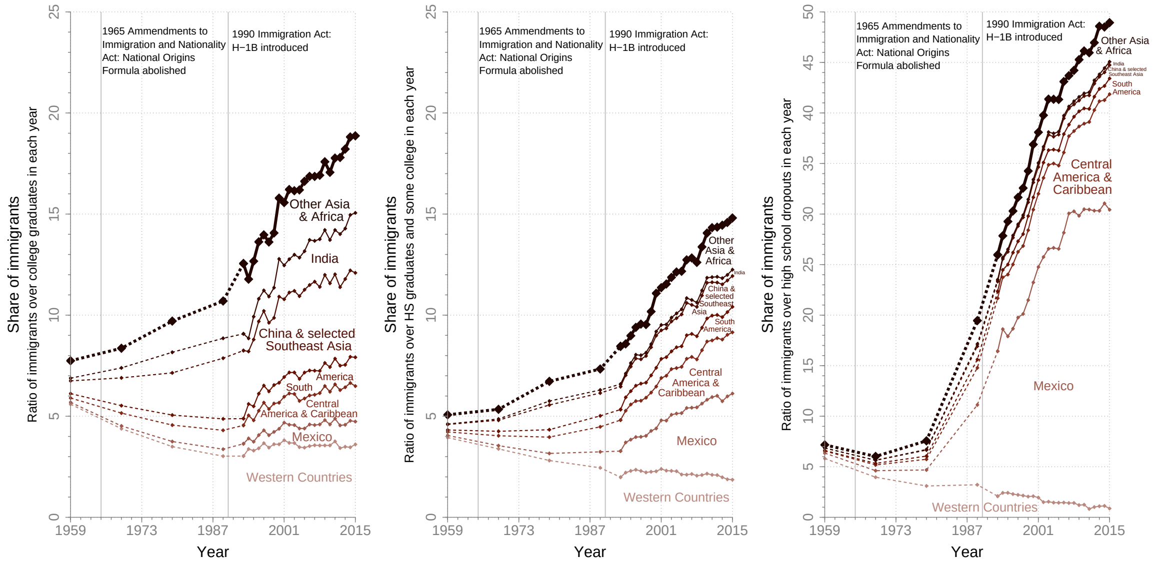
1.1 years for Western countries, Mexico, Central America and Caribbean, South America, China and selected Southeast Asia, India, and other Asia and Africa respectively.[12] Indian immigrants are the only group that increased education in 1990s by less than natives, but that is a special case, because they already had extremely high education levels to begin with.

Figure 2 further explores the importance of national origin distribution in understanding skill composition of immigration. Compared to Table 1, it provides a sense of the distribution of education by national origin, over and above averages. The figure shows that U.S. immigration is bimodal. In particular, they are mostly present among the highest educated (college graduates, 19% in 2015), and among the lowest educated (less than high school, 49%). It also shows that the relative importance of immigrants from each national origin in each of the education groups vary substantially. While Indian immigrants only represent 1.3% of the U.S. working-age population, they represent more than 3% of college graduates, and a negligible fraction of all individuals with less than high school. On the

[12] Some groups, like Western countries went from a slightly lower education level than natives until 1990 to higher level than them (with an increasing gap) after that.

Figure 2. Immigrant Share by National Origin and Education (1960-2015)

A. College graduates *B. HS graduates + some college* *C. Less than high school*



Note: Areas delimited by plotted lines show the share that immigrants from each national origin represent of all individuals aged over 24 with the indicated education. Country grouping and inter-census interpolations are described in Appendix A. *Sources:* Census data (1960-1990), and CPS (1994-2016).

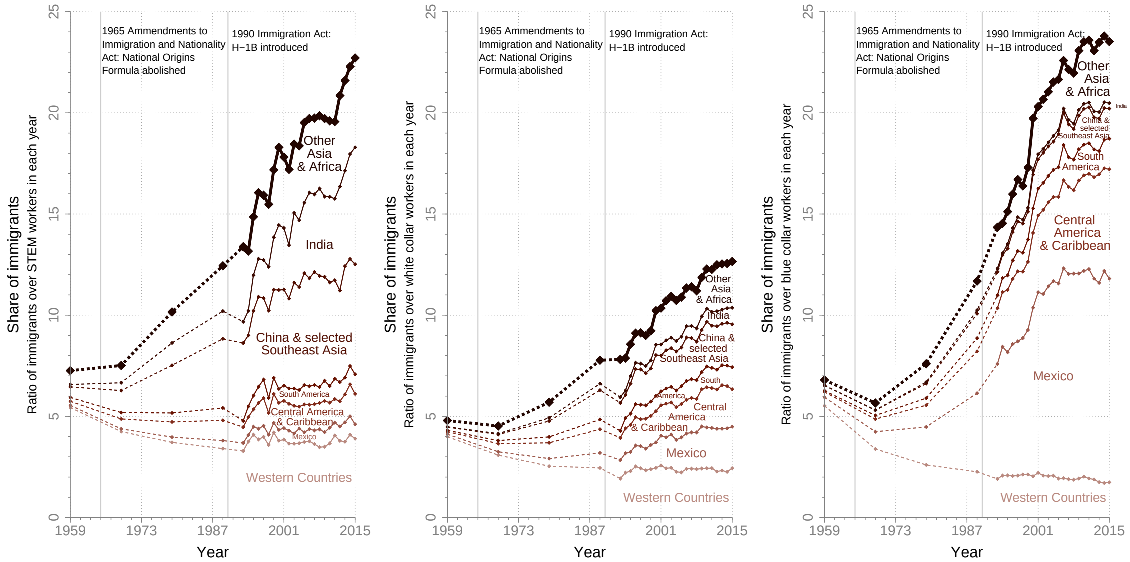
other extreme, while Mexican immigrants represent 5% of the U.S. population, and they represent 29% of all individuals with less than a high school diploma, they only represent slightly above 1% of all individuals with a college degree. Moreover, Panel A (and, importantly, not Panels B and C) also provides evidence of a change in the slopes after the introduction of H-1B visas in 1990.

Finally, Figure 3 provides a similar picture for occupations. While immigrants are relatively more resent in STEM and blue collar occupations, they are less present in white collar occupations. Mimicking the results for education, the vast majority of STEM occupations held by immigrants are executed by Asian nationals (more than 18.3% of all STEM employment, while they only represent less than 5% of the population), and very few of them are executed by Mexicans and Central Americans/Caribbeans (less than 3%, even though they represent more than 8% of the population). In this figure, the change in slope after the introduction of H-1B visas is more prominent, and it is mostly driven by Indian workers. This claim is clearly supported by more direct evidence since, for example, of all H-1B visas issued in 2016, 70.4% were issued to Indian nationals.[13] Overall, Indian workers represented 9.6% of STEM employment in 1993 and 18.3% in 2016.

[13] U.S. Department of State, Bureau of Consular Affairs, `https://travel.state.gov/content/dam/visas/Statistics/Non-Immigrant-Statistics/NIVDetailTables/FY16%20NIV%20Detail%20Table.xls`, accessed March 13th, 2017.

Figure 3. Immigrant Share by National Origin and Occupation (1960-2015)

A. STEM workers *B. White collar workers* *C. Blue collar workers*



Note: Areas delimited by plotted lines show the share that immigrants from each national origin represent of all individuals working in the indicated occupation. Country grouping and inter-census interpolations are described in Appendix A. *Sources:* Census data (1960-1990), and CPS (1994-2016).

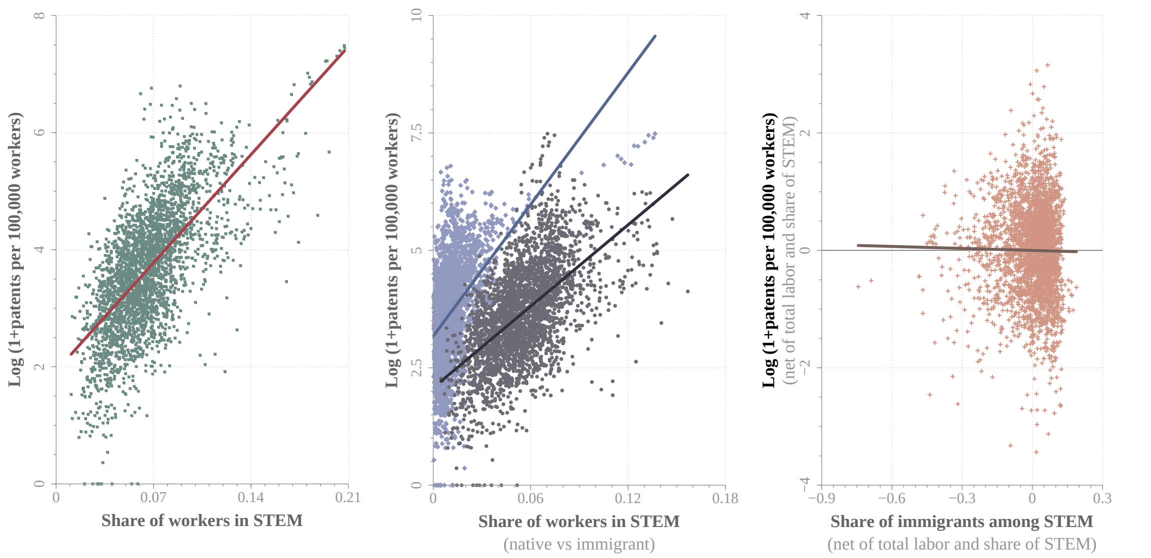
C. *Skilled Labor and the Generation of Ideas*

One of the central aspects of the model presented below is the presence of knowledge spillovers and externalities in the production of ideas from the use of STEM labor in production. To illustrate the extent of the association between STEM employment and the production of ideas, Figure 4 plots the spatial correlation between the (log of 1 plus) number of patents per 100,000 workers and different measurements of STEM intensity. To do so, I exploit metropolitan area variation in labor supply (obtained from the American Comunity Survey (ACS) for years 2000–2015) and in the number of patents (from the U.S. Patent and Trademark Office). In Panel A, STEM intensity is measured as the proportion of workers in the metropolitan area that are employed in STEM. The figure shows a very strong and positive correlation.

A relevant question for this paper is whether this correlation is driven by natives or immigrants. Panel B separates the STEM intensity in two parts: the one driven by natives, and the one driven by immigrants. In particular, the figure plots the share of all workers that are native (immigrant) STEM workers. The correlation stays strong and positive for both groups, even though regression lines seem to suggest a steeper pattern for natives. In order to corroborate or reject this different correlation, Panel C correlates the residuals of the regressions of the previous measure of patent productivity and the share of STEM workers that

Figure 4. Spatial Correlation Between STEM Labor and Patents (2000-2015)

A. Share of STEM workers *B. Native vs immigrant STEM* *C. Share of immigrants in STEM*



Note: The three figures plot the spatial correlation (scatter and regression fit) between different measures of relative STEM labor and the log of (one plus) the number of patents per 100,000 workers. The left figure measures STEM labor as the fraction of all workers that work in STEM. The central figure plots the share of all workers that are native STEM (purple/diamonds) and the share of all workers that are immigrant STEM (gray/circles). The right figure plots (the residuals of) the share of all STEM workers that are migrants (from a regression that controls for the number of workers in the metro area and the share of all workers that are STEM). An individual is defined as a worker if she worked at least 40 weeks in the reference year, and usually worked at least 20 hours per week. She is defined as a STEM worker if she worked in a STEM occupation and has a college degree. Immigrants are defined as foreign born individuals. *Sources:* ACS (2000–2015) for employment and U.S. Patent and Trademark Office for patents.

are immigrants on the total number of workers in the metropolitan area and the share of these workers that are employed in STEM occupations. The figure shows that, once labor market size and STEM intensity are controlled for, the correlation between immigrant intensity within STEM workers and the productivity in producing patents is essentially zero.

Overall, Figure 4 suggests a strong positive correlation between STEM intensity and productivity in patent production, and this correlation seems to be equally driven by native and immigrant STEM workers. As a final remark, it is important to note these correlations are only meant to show the link between STEM intensity and productivity in patent production, but the direction of causality could go in both directions. The structure of the model below provides a better framework to address these endogeneity concerns.

III. A Labor Market Equilibrium Model with Immigration and Knowledge Spillovers

In this section, I present a labor market equilibrium model with skilled and unskilled immigrants arriving from different countries of origin and competing with natives in three different occupations. The model, estimated with U.S. data, is then used to evaluate the selective immigration policies of interest, and to quantify the selectivity of immigration policy that maximizes native workers wellbeing. The modeling framework introduces heterogeneity across immigrants from different nationalities, accounts for human capital and labor supply adjustments by natives and previous generations of immigrants, and allows for economy-wide knowledge spillovers from skilled (STEM) workers and capital equipment. Through labor market competition, high skilled immigration reduces the incentives of natives to invest in human capital, and pushes them out of STEM occupations. Knowledge spillovers, however, may offset this competition effect inducing technological progress, which may be factor neutral or skill-biased. Given that firms do not internalize the effects of their input demands on knowledge spillovers, the use of STEM labor and capital equipment in production generates an externality. Immigration policy, thus, becomes a tool for policymakers to correct the underutilization of these inputs as a result of the externality.

A. Representative firm

A representative firm combines capital and labor to produce a single output. Knowledge spillovers are modeled as an externality through the production of ideas, measured as intellectual property products. Innovation is generated endogenously as a by-product of using STEM labor and capital equipment in general production. Replicating the behavior of a continuum of atomistic firms, the representative firm takes the stock of ideas as given, not internalizing the externality produced by equipment and STEM labor through knowledge spillovers.

To fix ideas, and provide some intuition on how innovation is generated in the model, consider the following example. When a company like Google creates a new technology –such Google Maps– it generates both marketable output –the Google Maps application– and new *knowledge* or ideas –smartphone-based digital mapping. This new knowledge can enable entirely new services developed by other firms.This knowledge can enable entirely new services developed by other firms. For example, digital mapping technologies made it possible for ride-hailing platforms such as Uber or Lyft to coordinate drivers and passengers in real time. Today these platforms operate with their own mapping technologies, and Google is not able to capture the value created by the subsequent applications of that knowledge.

Innovation. Let I_t denote the stock of ideas in period t, ΔI_t denote innovation (the net increase in the stock of ideas relative to the previous period), S_{Tt} denote the aggregate supply of STEM labor (measured in skill units), and K_{Et} denote equipment capital stock. Also let ξ_t denote an aggregate shock in the production of ideas. Innovation is generated when using equipment capital and STEM labor in general production, as described by the following technology:

$$\Delta I_t = \xi_t K_{Et}^{\chi_1} S_{Tt}^{\chi_2}. \tag{1}$$

The parameters χ_1 and χ_2 are not restricted to sum to one, thus allowing for increasing, constant, or decreasing returns to scale in the production of ideas. The exogenous innovation shock ξ_t is assumed to evolve according to:

$$\Delta \ln \xi_{t+1} = \pi_\xi + \sigma_\xi \upsilon_{\xi t+1}, \tag{2}$$

where $\upsilon_{\xi t}$ is a zero-mean innovation independently and identically distributed over time as a standard normal. The presence of such innovation shock is fundamental in identification, as discussed below, because it provides exogenous variation in innovation conditional on inputs.

Production function. Let K_{St} denote capital structures and S_{Bt} and S_{Wt} denote aggregate supplies of blue collar and white collar labor (skill units). Together with STEM labor S_{Tt} and equipment K_{Et}, they are the inputs in the production of aggregate output Y_t. Also let ζ_t denote an aggregate (factor-neutral) productivity shock which, combined with the stock of ideas determine total factor productivity. Output is produced according to the following nested CES technology:

$$Y_t = (\zeta_t I_r^{\varphi}) K_{St}^{\varsigma_t} \left\{ \alpha_t S_{Bt}^{\rho} + (1-\alpha_t) \left[\theta_t S_{Wt}^{\kappa} + (1-\theta_t) \left(\iota_t S_{Tt}^{\psi} + (1-\iota_t) K_{Et}^{\psi} \right)^{\frac{\kappa}{\psi}} \right]^{\frac{\rho}{\kappa}} \right\}^{\frac{1-\varsigma_t}{\rho}}, \tag{3}$$

where the demand shifters are allowed to evolve endogenously over time as a function of the relative importance of ideas in the economy:

$$o_t \equiv \frac{\exp\left(\tilde{o}_0 + \tilde{o}_1 \frac{I_t}{Y_t}\right)}{1 + \exp\left(\tilde{o}_0 + \tilde{o}_1 \frac{I_t}{Y_t}\right)} \qquad \text{for } o \in \{\varsigma, \alpha, \theta, \iota\}. \tag{4}$$

The inverse logit functional form ensures that the demand shifters lie between zero and one. The dependence on the ideas to output ratio, as opposed to the stock of ideas, prevents factor shares from tending to one or zero in the long run. The exogenous factor neutral productivity shock ζ_t evolves according to:

$$\Delta \ln \zeta_{t+1} = \pi_\zeta + \sigma_\zeta \upsilon_{\zeta t+1}, \tag{5}$$

where $\upsilon_{\zeta t+1}$ is a zero-mean innovation independently and identically distributed over time as a standard normal.

In this production function, innovation increases productivity and generates economic growth both in a factor neutral and in a skilled-biased way.[14] The term $\zeta_t I_r^\varphi$ can be interpreted as total factor productivity (TFP). To the extend to which $\varphi > 0$, the generation of ideas produces factor-neutral technological progress by enhancing TFP in the spirit of Romer (1986) and Lucas (1988). Furthermore, innovation also shifts the relative demands of inputs by changing ς_t, α_t, θ_t, and/or ι_t, thus inducing endogenous skill-biased technological change (as long as $\tilde{o}_1 \neq 0$ for some $o \in \{\varsigma, \alpha, \theta, \iota\}$). For example, the invention of computers may foster economic growth (TFP), increase the relative productivity of STEM labor, and/or substitute out blue or white collar labor. As in Krusell et al. (2000), this production function can also produce skilled-biased technical change through capital-skill complementarity as a result of an exogenous fall in the prices of equipment.[15]

Related to other production functions specified in the immigration literature, this production function extends the one used in Llull (2018a, 2021) to allow for a third input (STEM labor) and to incorporate knowledge spillovers in the way described above. It further differs from the nested CES structure introduced in the immigration literature by Borjas (2003) and Ottaviano and Peri (2012). Unlike in these papers, the explicit modeling of labor supply allows accounting for occupational decisions of immigrants, which itself generates, endogenously, the imperfect substitutability estimated in the literature. Specifically, Ottaviano and Peri (2012) discuss the importance of imperfect substitutability between natives and immigrants with the same observable skills "because they tend to work in different occupations". The equilibrium structure generates this imperfect substitutability (even though immigrants within a given occupation are perfect substitutes) through endogenous sorting into occupations. In fact, Llull (2018a) shows that a similar model generates a "reduced form" elasticity of substitution between na-

[14] The introduction of intangible capital in the aggregate production function has already been discussed in the macroeconomics literature (e.g. McGrattan and Prescott, 2010, 2014). These papers, however, model it as a (rival) intermediate input in which the firm invests, not as a non-rival stock of ideas generated unintendedly in general production.

[15] Lewis (2011, 2013) discusses the importance of capital-skill complementarity in measuring labor market effects of immigration.

tives and immigrants that fits well within the ballpark of estimates in Ottaviano and Peri (2012). Albert, Glitz and Llull (2026) are also able to endogenously generate a similar "reduced form" elasticity using a different production function.

Profit maximization. The representative firm maximizes profits in a static way. Given the single output production, I normalize output prices to one. Let r_{jt} for $j \in \{T, W, B\}$ denote the returns paid by the firm to STEM, white collar, and blue collar labor skill units. Let r_{St} and r_{Et} denote the rental rates rates paid for structures and equipment capital respectively. The firm's problem is given by:

$$\max_{S_{Tt}, S_{Wt}, S_{B_t}, K_{Et}, K_{St}} \left\{ \begin{array}{c} Y(\zeta_t, I_t, S_{Tt}, S_{Wt}, S_{B_t}, K_{Et}, K_{St}) - r_{Tt} S_{Tt} \\ -r_{Wt} S_{Wt} - r_{Bt} S_{Bt} - r_{Et} K_{Et} - r_{St} K_{St} \end{array} \right\}. \tag{6}$$

The representative firm takes the aggregate stock of ideas I_t as given. This reflects that the contribution of each atomistic firm to it is very small. Given that ideas are non-rival, firms do not internalize the effect that their inputs demands have on the aggregate production of ideas, which generates, as a result, an externality.

Define the following Armington aggregators for the different parts of the CES production function: $Q_{1t} \equiv (\iota_t S_{Tt}^{\psi} + (1 - \iota_t) K_{Et}^{\psi})^{1/\psi}$, $Q_{2t} \equiv (\theta_t S_{Wt}^{\kappa} + (1 - \theta_t) Q_{1t}^{\kappa})^{1/\kappa}$, and $Q_{3t} \equiv (\alpha_t S_{Bt}^{\rho} + (1 - \alpha_t) Q_{2t}^{\rho})^{1/\rho}$. The aggregate demand of STEM skill units is:

$$r_{Tt} = (1 - \varsigma_t)(1 - \alpha_t)(1 - \theta_t)\iota_t \left(\frac{Q_{2t}}{Q_{3t}}\right)^{\rho} \left(\frac{Q_{1_t}}{Q_{2t}}\right)^{\kappa} \left(\frac{S_{Tt}}{Q_{1t}}\right)^{\psi} \frac{Y_t}{S_{Tt}}. \tag{7}$$

The demand of white collar skill units is given by:

$$r_{Wt} = (1 - \varsigma_t)(1 - \alpha_t)\theta_t \left(\frac{Q_{2t}}{Q_{3t}}\right)^{\rho} \left(\frac{S_{Wt}}{Q_{2t}}\right)^{\kappa} \frac{Y_t}{S_{Wt}}. \tag{8}$$

The demand of of blue collar skill units is given by:

$$r_{Bt} = (1 - \varsigma_t)\alpha_t \left(\frac{S_{Bt}}{Q_{3t}}\right)^{\rho} \frac{Y_t}{S_{Bt}}. \tag{9}$$

Finally, the demands for capital structures and equipment are:

$$r_{St} = \varsigma_t \frac{Y_t}{K_{St}}, \tag{10}$$

and:

$$r_{Et} = (1 - \varsigma_t)(1 - \alpha_t)(1 - \theta_t)(1 - \iota_t) \left(\frac{Q_{2t}}{Q_{3t}}\right)^{\rho} \left(\frac{Q_{1_t}}{Q_{2t}}\right)^{\kappa} \left(\frac{K_{Et}}{Q_{1t}}\right)^{\psi} \frac{Y_t}{K_{Et}}. \tag{11}$$

It is important to note that, despite the externalities, the representative firm makes zero profits, consistent with free entry of atomistic firms. From the point of view of the atomistic firms, the above production function is, thus, constant returns to scale, even if $\varphi > 0$ or $\tilde{o}_1 \neq 0$ for some $o \in \{\varsigma, \alpha, \theta, \iota\}$, given that factor shares sum to one, as it can be trivially shown from (7) through (11).

B. Workers

Workers make life-cycle decisions on education, occupation and participation. Consistent with standard models of human capital (Ben-Porath, 1967) they concentrate education at the beginning of their careers, and then keep accumulating human capital in the form of experience throughout their working life. Given the dynamics of the model, individuals face the trade-off between enjoying wages/utility today and investing in human capital for the future. In equilibrium, immigration affects this trade-off by changing relative wages.

Wages are specified in a Mincerian way (Mincer, 1974) as the product of skill units and their market price. This specification implies a direct and an indirect effect of immigration on wages. The direct effect is through skill prices. Immigration affects relative skill prices in equilibrium due to the change in relative supplies and potential knowledge spillovers it induces. Indirectly, immigration affects wages by affecting native decisions (and, as a result, their skill units).

The model allows natives to adjust to immigration in differently at different points of the life cycle. Young individuals may be more likely to change occupations and, if still in school, to adjust their education decision, while older individuals, who devoted all their career to a given occupation, may exhibit lower occupational mobility. The labor supply decision problem is defined by a dynamic discrete choice model similar to Keane and Wolpin (1994, 1997), Lee (2005), Lee and Wolpin (2006, 2010), and Llull (2018a, 2021). Unlike these models, decisions are modeled to be of nested-logit type, whereas wages are kept log-normal as a standard Roy (1951) model. Similar to Heckman et al. (1998), the model abstracts from accumulating occupation-specific work experience, even though it allows by age-specific transition costs that capture the different life-cycle propensities to adjust. As it is the case in Lee and Wolpin (2006, 2010) and Llull (2018a, 2021), and similar to the spirit of Altuğ and Miller (1998) and Krusell and Smith Jr. (1998), the combination of aggregate and idiosyncratic shocks requires an approximation to rational expectations, discussed below.

The model features a substantial amount of (observed) heterogeneity. Besides realism, this heterogeneity allows me to better capture the composition of immigrant inflows (see Section II.B) and their choices. Some of the heterogeneity (e.g. preschool children) provide exclusion restrictions to identify the parameters of the model as discussed below. The model also captures important features discussed in the immigration literature. Endogenously, through occupational choices, immigrants are allowed to downgrade at arrival (Dustmann et al., 2013). The model also features wage assimilation: among two individuals with the same observable skills, the one who spend more time in the U.S. earns more given the higher return to domestic experience than to foreign one (LaLonde and Topel, 1992). The labor market competition channel for assimilation, analyzed in Albert, Glitz and Llull (2026), is also present in the model, because natives and immigrants, and immigrants with different time in the U.S. tend to work in different occupations, which makes them imperfect substitutes in production (Ottaviano and Peri, 2012).

Life cycle. Let $a \in \{16, ..., 65\}$ denote age. Native individuals are born with $a = 16$ and a given initial human capital endowment, to be specified below. They make yearly decisions until $a = 65$, when they die with certainty. Immigrants only start making decisions upon entry in the United States, which occurs at a given (individual-specific and exogenous) age of entry $\tilde{a}$. There is no return migration, so they make yearly endogenous decisions over ages $a \in \{\tilde{a}, ..., 65\}$.

Choice sets. Every year, individuals decide one of three to five mutually exclusive alternatives: working in blue collar, white collar, or STEM, attending school, and staying home. The STEM occupation has a college degree requirement, and therefore is not available to individuals with less than 16 years of education. Likewise, consistently with the very low rates of school reentry observed in the data, I assume that leaving school is an absorbing state. Thus, school is only available as a choice if previous decision was to attend school. As a result, the choice set depends on previous choice and education level. Denote the choice set as $\mathcal{D}(h_a)$, where h_a is the vector of individual-specific state variables defined below, which includes education and previous period decision among other variables. The choice of a given individual at age a is denoted by $d_a \in \mathcal{D}(h_a)$, and a set of indicator variables is defined, such that $d_{ja} \equiv \mathbb{1}\{d_a = j\}$ for any $j \in \mathcal{D}(h_a)$, with $\sum_{j \in \mathcal{D}(h_a)} d_{ja} = 1$.[16]

Observable idiosyncratic state variables. Let $h_a \equiv (a, \ell, E_a, d_{a-1}, n_a, \tilde{a})'$ denote the vector of observable (by the econometrician) individual-specific state variables. This vector is defined by age a, demographic type ℓ, education E_a, lagged choice d_{a-1}, number of preschool children at home $n_a \in \mathcal{C} \equiv \{0, 1, 2+\}$, and, in the case of immigrants, age at entry $\tilde{a}$. I partition the workforce into a finite number of types, subscripted by ℓ, defined by observable characteristics. Na-

[16] The indicator function $\mathbb{1}\{\cdot\}$ is defined to be one if the argument is satisfied, zero otherwise.

tives are classified into six groups, defined by race (Hispanic, non-Hispanic black, non-Hispanic non-black) and sex (male and female). Immigrants are divided into fourteen groups, defined by national origin (Western countries, Mexico, Central America & Caribbean, South America, China & selected Southeast Asia, India, and other Asia and Africa) and sex. These groups identify twenty types of individuals, $\mathcal{L} \equiv (\mathcal{R} \times \mathcal{G}) \cup (\mathcal{O} \times \mathcal{G})$, where $\mathcal{R}$ denotes the set of races for natives, $\mathcal{O}$ denotes the set of national origins of immigrants, and $\mathcal{G} \equiv \{1, 2\}$ denotes the set of sexes (male and female are denoted by 1 and 2 respectively). I also denote by $\tilde{\mathcal{L}}_{\tilde{\ell}(\ell)}$ four sets of combined types, which, indexed by $\tilde{\ell}(\ell) \in \{1, 2, 3, 4\}$, define native male $\tilde{\mathcal{L}}_1 \equiv \{\ell : \ell \in (\mathcal{R} \times \{1\})\}$, native female $\tilde{\mathcal{L}}_2 \equiv \{\ell : \ell \in (\mathcal{R} \times \{2\})\}$, immigrant male $\tilde{\mathcal{L}}_3 \equiv \{\ell : \ell \in (\mathcal{O} \times \{1\})\}$, and immigrant female $\tilde{\mathcal{L}}_4 \equiv \{\ell : \ell \in (\mathcal{O} \times \{2\})\}$.

The initial state vector for natives is given by $h_{16} = (16, \ell, E_{16}, d_{15}, 0, \cdot)'$, where ℓ and E_{16} are exogenously determined at birth, and $d_{15} = S$ if $E_{16} = 12$ and $d_{15} = H$ otherwise. Immigrants enter into the United States with a given state vector $h_{\tilde{a}} = (\tilde{a}, \ell, E_{\tilde{a}}, d_{\tilde{a}-1}, 0, \tilde{a})'$, where $d_{\tilde{a}-1} = S$ if $E_{\tilde{a}} \geq (\tilde{a} - 6) - 2$, and $d_{\tilde{a}-1} \equiv F$ otherwise.[17] The distribution of initial state variables of natives and immigrants is specified outside of the model. The state vector is updated as follows: a is increased in one unit, ℓ is constant, $E_{a+1} = E_a + d_{Sa}$, the previous period choice d_{a-1} is replaced by the current choice d_a, $\tilde{a}$ is constant, and the number of children is increased stochastically with cumulative distribution function $P_n(n|h_a, d_a)$ with $P_n(n|h_a, j) = P_n(n|h_a, k)$ for any $j, k \neq S$.[18] The set of possible values for the observable idiosyncratic state variables at age $a+l$ when the state variable was h_a at age a is denoted by $\mathcal{H}_{a+l|h_a}$, and the unconditional set of possible values is $\mathcal{H}$.

Idiosyncratic shocks. This model includes two types of idiosyncratic shocks, which are independent and identically distributed across individuals and over time. Let η_a denote a shock to individual productivity, with $\eta_a | h_a \sim \mathcal{N}(0, 1)$. Also let ϵ_{ja} denote a taste shock associated to alternative $j \in \mathcal{D}(h_a)$. I define the vector of convoluted idiosyncratic shocks, denoted by ε_a, as:

$$\varepsilon_a \equiv (\sigma_{B\ell}\Lambda_0\eta_a + \epsilon_{Ba}, \sigma_{W\ell}\Lambda_0\eta_a + \epsilon_{Wa}, \sigma_{T\ell}\Lambda_0\eta_a + \epsilon_{Ta}, \epsilon_{Sa}, \epsilon_{Ha})', \tag{12}$$

[17] This assumption implies that immigrants with up to two years of work experience abroad can still enroll in school when they arrive in the United States. Given data availability, I assume that $F = H$, so that the cost of reentry to work is the same if the individual was in the U.S. but not working or she was abroad.

[18] In order to capture the demographic transition, and the subsequent increase in female labor force participation, I assume that the transition function before 1970 was $\tilde{P}_n(n|h_a, d_a)$ instead of $P_n(n|h_a, d_a)$. For tractability, I assume that the affected cohorts do not take into account the change in the fertility process when, before the change, form expectations about the future.

with $\sigma_{j\ell}$ and Λ_0 defined below. The distribution of ϵ_a is such that the combined idiosyncratic shock is generalized extreme value distributed, $\varepsilon_a|h_a \sim F_\varepsilon(\varepsilon_a)$ with:

$$F_\varepsilon(\varepsilon_a) \equiv \exp\left\{-\left[\left(e^{-\varepsilon_{Ba}/\varrho} + e^{-\varepsilon_{Wa}/\varrho} + e^{-\varepsilon_{Ta}/\varrho}\right)^\varrho + e^{-\varepsilon_{Sa}} + e^{-\varepsilon_{Ha}}\right]\right\}, \tag{13}$$

where $\varrho \equiv \sqrt{1 - \text{Corr}(\varepsilon_{ja}, \varepsilon_{ka})}$ for any $j, k \in \{B, W, T\}$.[19]

Wages, skill units, and aggregate state variables. Individual wages in this model are defined as the product of the amount of skill units supplied by the individual in occupation j, and the market price of these skill units. Let r_t denote the vector of skill prices at time t, defined as $r_t \equiv (r_{Bt}, r_{Wt}, r_{Tt})'$. Let $s_j(h_a, \eta_a)$ denote the amount of skill units supplied by an individual with idiosyncratic state variables h_a and η_a. The occupation-j wage of this individual is defined as:

$$w_j(h_a, \eta_a, r_t) \equiv r_{jt} s_j(h_a, \eta_a). \tag{14}$$

As key determinants of individual choices, aggregate skill prices r_t are state variables. Given the presence of aggregate shocks (and uncertain future migration, as discussed below), individuals cannot perfectly predict future skill prices $\{r_{t+l}\}_{l\in\{1,\ldots,65-a\}}$. Let ϖ_t denote all information available to individuals at time t to forecast them. The state vector for the worker decision problem is thus expanded to also include ϖ_t. What is included in ϖ_t is discussed below.

I assume skill units $s_j(h_a, \eta_a)$ are log-additively separable:

$$s_j(h_a, \eta_a) \equiv \exp(\tilde{s}_j(h_a) + \sigma_{j\ell}\eta_a), \tag{15}$$

where $\tilde{s}_j(\cdot)$ is a function of the observable idiosyncratic state vector h_a, and $\sigma^2_{j\ell}$ is the occupation-j-type-ℓ-specific conditional variance. I assume that the fifth element of h_a (number of children) does not enter $\tilde{s}_j(\cdot)$, which provides an exclusion restriction for identification, as discussed below.

The skill units production function, specified in Equation (15) allows for different transition costs at different ages through the interaction of previous period choice and age. Furthermore, the error structure (along with these costs) gives freedom to the model to fit at the same time the persistence in choices and wages in the data and the observed variance in wages. Correctly reproducing the transition probability across alternatives is crucial to credibly identify labor supply and human capital adjustments to immigration.

[19] Appendix B characterizes the distribution of ϵ_a and provides a discussion of how to numerically simulate productivity and taste shocks.

Alternative-specific period utility functions. I assume that individuals are not allowed to save or borrow. Individuals thus consume all earned income when they work (wages). Consumption utility in non-working alternatives is assumed to be policy-invariant and embedded in model parameters. The utility of consumption is logarithmic, and other non-pecuniary elements enter additively. Consumption utility is scaled by a parameter Λ_0, which captures how individuals value consumption relative to the amenity.

Non-pecuniary utility associated to working in occupation j is given by a type-specific amenity value, denoted by $\Lambda_{1j}(\ell)$, an occupation-specific re-entry cost if the individual was at home in the previous period Λ_{2j}, and the occupation-specific taste-shock ϵ_{ja}, defined above:

$$\begin{aligned} u_j(h_a, \varepsilon_a, r_t) &\equiv \Lambda_0 \ln w_j(h_a, \eta_a, r_t) + \Lambda_{1j}(\ell) - \Lambda_{2j} d_{Ha-1} + \epsilon_{ja} \\ &= \Lambda_0[\ln r_{jt} + \tilde{s}_j(h_a)] + \Lambda_{1j}(\ell) - \Lambda_{2j} d_{Ha-1} + \varepsilon_{ja}, \quad j \in \{B, W, T\}, \end{aligned} \tag{16}$$

The net utility of attending school is assumed to depend on individual type ℓ, educational level E_a, and the idiosyncratic taste shock ε_{Sa}:

$$u_S(h_a, \varepsilon_a, r_t) \equiv \tau(\ell, E_a) + \varepsilon_{Sa}. \tag{17}$$

Finally, the utility of staying home depends on individual type, the number of children n_a, and the taste shock ε_{Ha}:

$$u_H(h_a, \varepsilon_a, r_t) \equiv \vartheta(\ell, n_a) + \varepsilon_{Ha}. \tag{18}$$

Intertemporal decisions. Let β denote the subjective discount factor. Also let $\tilde{u}_j(\cdot)$ denote the deterministic part of the period utility function, defined as $\tilde{u}_j(h_a, r_t) \equiv u_j(h_a, \varepsilon_a, r_t) - \varepsilon_{ja}$.[20] An individual with a state vector h_a and idiosyncratic shock ε_a, observed at time t (that is, when equilibrium prices are r_t and the information available to predict their future values is ϖ_t), chooses $\{d_{ja}\}_{j \in \mathcal{D}(h_a)}$ to sequentially maximize the expected discounted sum of payoffs:

$$\mathbb{E}_t \left[\sum_{l=0}^{65-a} \beta^l \left(\sum_{j \in \mathcal{D}(h_{a+l})} d_{ja+l} [\tilde{u}_j(h_{a+l}, r_{t+l}) + \varepsilon_{ja+l}] \right) \right], \tag{19}$$

where $\mathbb{E}_t[\,\cdot\,] \equiv \mathbb{E}[\,\cdot\,|h_a, \varepsilon_a, r_t, \varpi_t]$ denotes the conditional expectation of the argument given the information available to the individual at time t, including h_a,

[20] Given that ε_{ja} enters $u_j(\cdot)$ additively, and that ε_{ka} for any $k \neq j$ does not enter $u_j(\cdot)$, the above expression does not depend on ε_a.

ε_a, r_t, and ϖ_t. Appealing to the Bellman's principle (Bellman, 1957), I express worker's decision problem recursively as:

$$V(h_a, \varepsilon_a, r_t, \varpi_t) = \max_{\{d_{ja}\}_{j\in\mathcal{D}(h_a)}} \sum_{j\in\mathcal{D}(h_a)} d_{ja} \left\{\tilde{u}_j(h_a, r_t) + \varepsilon_{ja} + \beta\, \mathbb{E}_t[V(h_{a+1}, \varepsilon_{a+1}, r_{t+1}, \varpi_{t+1})]\right\}, \tag{20}$$

where the terminal value is defined to be zero, $V(h_{65}, \varepsilon_{65}, r_t, \varpi_t) \equiv 0$. Following Arcidiacono and Miller (2011), I define two additional objects derived from the above expression. First, define the ex-ante value function (the continuation value of being in state (h_a, r_t, ϖ_t), just before ε_a is revealed) as:

$$\bar{V}(h_a, r_t, \varpi_t) \equiv \int V(h_a, \varepsilon, r_t, \varpi_t) dF_\varepsilon(\varepsilon). \tag{21}$$

Second, define the alternative-specific conditional value function as:

$$v_j(h_a, r_t, \varpi_t) \equiv \tilde{u}_j(h_a, r_t) + \beta \int \sum_{h\in\mathcal{H}_{a+1|h_a}} \bar{V}(h, r, \varpi) P_h(h|h_a, j) dF_r(r, \varpi|\varpi_t, r_t), \tag{22}$$

where the function $F_r(r, \varpi|\varpi_t, r_t)$ is the distribution of aggregate conditions in period $t+1$ given information available at time t, and $P_h(h|h_a, j)$ is the transition probability mass function for h_a discussed above, which is degenerate for all elements of h except for n. Thus, optimal choices, denoted by $d^*_{ja}(h_a, \varepsilon_a, r_t, \varpi_t)$ for $j \in \mathcal{D}(h_a)$, are given by:

$$\{d^*_{ja}(h_a, \varepsilon_a, r_t, \varpi_t)\}_{j\in\mathcal{D}(h_a)} = \arg\max_{\{d_{ja}\}_{j\in\mathcal{D}(h_a)}} \sum_{j\in\mathcal{D}(h_a)} d_{ja}[v_j(h_a, r_t, \varpi_t) + \varepsilon_{ja}]. \tag{23}$$

Given the specific distribution for ε introduced above the solution to this implies:

$$\bar{V}(\cdot) = \gamma + \ln\left[\left(e^{v_B(\cdot)/\varrho} + e^{v_W(\cdot)/\varrho} + e^{v_T(\cdot)/\varrho}\right)^{\varrho} + e^{v_S(\cdot)} + e^{v_H(\cdot)}\right]. \tag{24}$$

where γ is the Euler's constant, approximately equal to 0.5772.

Expectations Rational individuals use the true distributions of stochastic state variables $F_\varepsilon(\varepsilon_a)$, $P_h(h_{a+1}|h_a, d_a)$, and $F_r(r_{t+1}|r_t, \varpi_t)$ to form expectations about the future. All these functions are explicitly specified in the model except for $F_r(r_{t+1}|r_t, \varpi_t)$. Given the presence of aggregate and idiosyncratic shocks, rational expectations imply that ϖ_t includes the entire distribution of state variables in period t, which is intractable.[21] This is a well known problem in macroeconomics and applied microeconometrics that has been addressed by approximating

[21] Despite this complication, aggregate shocks are very necessary in the immigration context to avoid making the unrealistic assumption that workers can perfectly predict the future evolution of economic conditions including the future inflow of migrants.

rational expectations by simpler forecasting rules based on equilibrium outcomes (Krusell and Smith Jr., 1998; Altuğ and Miller, 1998; Lee and Wolpin, 2006, 2010; Llull, 2018a, 2021). In the immigration context, Llull (2018a, 2021) finds that an autoregressive process in first differences and the contemporaneous change in the aggregate shock can explain 99.9% of the variation in the level of skill prices. Further results presented in Llull (2018b) show that the innovation in the aggregate shock alone can still explain 99.9% of the variation in levels (along with current skill price), and 71–76% of the first differences. Given this, and following a similar approach as in Altuğ and Miller (1998), I assume skill prices are forecasted using the following rule:

$$\ln r_{jt+1} = \ln r_{jt} + \Xi_{0j} + \Xi_{1j}\sigma_{\zeta}\upsilon_{\zeta t+1} + \Xi_{2j}\sigma_{\xi}\upsilon_{\xi t+1} + \sigma_{\Upsilon j}\Upsilon_{jt+1}, \quad j \in \{B, W, T\}, \tag{25}$$

where Υ_{jt+1} is an approximation error uncorrelated with the ideas and TFP shocks $\sigma_{\xi}\upsilon_{\xi t+1}$ and $\sigma_{\zeta}\upsilon_{\zeta t+1}$, and Ξ_{0j}, Ξ_{1j}, Ξ_{2j}, and $\sigma_{\Upsilon j}$ are not parameters, but, instead, implicit functions of the fundamentals, and, hence, part of the solution of the model. Equation (25) implicitly assumes that r_t is a sufficient statistic of all the information that individuals have at time t to predict r_{t+1} (as $\upsilon_{\zeta t+1}$ and $\upsilon_{\xi t+1}$ are unknown at t and i.i.d. over time). This implies ϖ_t is a redundant state variable and, thus, I omit it hereinafter.

C. Capitalists

The discussion in Borjas (2003) and Ottaviano and Peri (2012) suggests that, even though the model described so far is informative on distributional effects of immigration, the overall labor market effects of immigration depend on our assumption of how capital reacts to immigration. These papers take one of the two most extreme assumptions: Borjas (2003) assumes that capital does not adjust to immigration, whereas in Ottaviano and Peri (2012), interest rates do not react (long run small open economy). Llull (2018a) provides results with both assumptions noting that reality should probably be somewhere in between. Llull (2021) takes a simplified equilibrium approach, allowing capital to adjust assuming that capital supply is given by a reduced form λK capital supply, following the theoretical specification in Dustmann et al. (2016).

In this paper, I close the economy with a structural specification of capital supply that allows me to simulate general equilibrium effects. In particular, I assume that capital is supplied by a continuum of infinitely lived capitalists that make consumption and savings decisions, and live out of the return to their assets. These capitalists are homogeneous, and therefore are characterized by a representative consumer model.

Let A_t denote the initial assets position of these capitalists in year t, decided in period $t-1$. At the end of period $t-1$, these assets are transformed into capital and rented to the firms, as discussed below. After production takes place, capitalists receive back their assets plus a return r_{At}. After returns are paid, the individual decides how much to consume, C_t, and how much to save for the next period, A_{t+1}. The problem of the representative capitalist is given by:

$$\max_{\{C_{t+\tau},A_{t+1+\tau}\}_{\tau=0}^{\infty}} \sum_{\tau=0}^{\infty} \beta^{\tau}\, \mathbb{E}_t \left[\frac{C_{t+\tau}^{1-\nu}}{1-\nu} \right], \quad \nu > 0 \tag{26}$$

subject to:

$$C_{t+\tau} + A_{t+1+\tau} \leq (1 + r_{At+\tau})A_{t+\tau}, \tag{27}$$

$$A_{t+1+\tau} \geq 0. \tag{28}$$

The first order conditions of this problem yield the standard Euler equation:

$$C_t^{-\nu} = \beta\, \mathbb{E}_t[(1 + r_{At+1})C_{t+1}^{-\nu}]. \tag{29}$$

Given that equilibrium asset returns also depend on labor supply, which is determined by aggregate and idiosyncratic shocks, the Krusell and Smith Jr. (1998) problem discussed above for labor also applies to capital. Following a similar logic, I assume capitalists form expectations about future returns to assets using the following approximation rule:

$$\ln(1 + r_{A,t+1}) = \ln(1 + r_{At}) + \Xi_{0A} + \Xi_{1A}\sigma_{\zeta}\upsilon_{\zeta,t+1} + \Xi_{2A}\sigma_{\xi}\upsilon_{\xi,t+1} + \sigma_{\Upsilon A}\Upsilon_{A,t+1}, \tag{30}$$

where $\Upsilon_{A,t+1} \sim N(0,1)$ is an approximation error uncorrelated with the ideas and TFP shocks $\sigma_{\zeta}\upsilon_{\zeta,t+1}$ and $\sigma_{\xi}\upsilon_{\xi,t+1}$, and Ξ_{0A}, Ξ_{1A}, Ξ_{2A}, and $\sigma_{\Upsilon A}$ are not parameters, but, instead, implicit functions of the fundamentals, and, hence, part of the solution of the model. As in the workers' problem, this forecasting rule implicitly assumes that current aggregate returns are a sufficient statistic for the information available at time t to predict future returns.

Given the assumptions above (CRRA preferences, linear budget set, and expectation rule), assets accumulation are given by:

$$A_{t+1} = \mathbb{s}(r_{At}; \beta, \nu)(1 + r_{At})A_t, \tag{31}$$

where $\mathbb{s}(\cdot)$ is the policy function of the capitalists, expressed in as a savings rate.

There is not a perfect mapping between assets and capital in this model, because assets can be transformed into the two types of capital goods at different relative prices. In particular, one unit of structures is normalized to cost one unit of assets, while one unit of equipment costs q_{Et} units of assets.[22] These prices are taken as exogenous in the model.

In the model there is a competitive zero-profit intermediary that, at the end of period $t-1$ transforms assets A_t into capital goods. These capital goods are rented to the firm for production in period t. At the end of the period, the intermediary and returns to the capitalists the (asset-transformed) undepreciated capital plus rental income. The intermediary allocates assets across structures (K_{St}) and equipment (K_{Et}) subject to the feasibility constraint:

$$A_t \geq K_{St} + q_{Et-1} K_{Et}. \tag{32}$$

At the end of the period, the intermediary receives rental payments from firms and recovers the undepreciated capital. Because the relative price of equipment may change over time, the remaining undepreciated capital and the rental returns are transformed back into assets at prices q_{Et}. The asset value of the intermediary's portfolio at the end of the period is therefore:

$$(1 + r_{At})A_t = (1 - \delta_{St} + r_{St})K_{St} + \left[r_{Et} + (1 - \delta_{Et})q_{Et}\right] K_{Et}, \tag{33}$$

where δ_{St} and δ_{Et} denote the depreciation rates of structures and equipment.

Since the intermediary operates a constant-returns technology under free entry, an interior allocation across the two capital types requires equalized returns per unit of assets invested. The no-arbitrage condition therefore implies:

$$1 + r_{At} = 1 - \delta_{St} + r_{St} = \frac{r_{Et} + (1 - \delta_{Et})q_{Et}}{q_{Et-1}}. \tag{34}$$

If this condition does not hold, the intermediary allocates all assets to the capital type offering the highest return.

D. Equilibrium

The market structure in this model is as follows. Immigrant inflows are specified outside of the model. Interest rates are such that capital markets clear. Returns to skill units are such that labor supply equals labor demand.

[22] Krusell et al. (2000) link the fall in equipment prices to the increase in the college-high school wage gap. It is important to keep this ingredient in the model so that it has room for exogenous skill-biased technical change not driven by the accumulation of ideas.

The equilibrium capital is determined as follows. Only interior solutions of the intermediary's problem are considered. This means that equilibrium returns to capital satisfy (34). Replacing (10) and (11) into (34) determines the relative capital supply in equilibrium of each of the two types of capital given the other inputs. Intermediaries invest all assets in each period, which implies that (32) is satisfied with equality. Combined with the previous expression, this determines the mapping between assets and each type of capital. Given an initially predetermined level of assets A_0, the law of motion of capital (31) determines capital supplies in equilibrium.

The labor market prices of skill units are also determined by market clearing conditions. Aggregate supply of skills in occupation $j \in \{B, W, T\}$, denoted by $S_{jt}^{(S)}(r_t)$, is given by the aggregation over all skill units supplied by individuals working in occupation j:

$$S_{jt}^{(S)}(r_t) = \iint \sum_{h \in \mathcal{H}} d_{jt}^{*}(h, \epsilon + \eta, r_t) s_j(h, \eta) P_h(h) dF_{\epsilon}(\epsilon) d\Phi(\eta), \tag{35}$$

where $P_h(\cdot)$ is the probability mass function for each point of the (idiosyncratic) state space, $\Phi(\cdot)$ denotes the standard normal cumulative distribution function, and $F_{\epsilon}(\cdot)$ is the cumulative distribution function of the taste shock ϵ_a.[23] Aggregate labor demands, denoted by $S_{jt}^{(D)}(r_t)$ for $j \in \{B, W, T\}$, are given by the solution of the system of equations defined by (7) through (9), replacing capital by the equilibrium conditions described above.

Even though migration decisions are not modeled, immigration inflows (both size and composition) and capital supplies are endogenously determined by processes specified outside of the model. This is so because, as noted below, no orthogonality condition is assumed between the aggregate productivity shocks, ζ_t and ξ_t, and the aggregate variables (migration process, cohort sizes,...). Thus, immigrant inflows are allowed to react to changes in the economic conditions in the U.S. and other aggregate factors, like endogenous immigration policies.[24]

IV. Identification

The subjective discount factor β is assumed to be equal to 0.95. The relative price of equipment capital q_{Et} and capital depreciation rates, δ_{Et} and δ_{St}, are directly observed in the data. The transition function for preschool children $P_n(\cdot)$

[23] Equation (35) assumes that there is a measure 1 of workers. In the empirical application, this measure is scaled by population size.

[24] Llull (2018a) provides evidence that, in equilibrium, aggregate flows seem to correlate with aggregate shocks, while composition remains rather invariant. The counterfactual evolution of these variables are precisely determined by the design of the policy experiments simulated below.

is directly identified from observed transitions in census microdata. In line with the literature, and as specified above, the distributions of idiosyncratic and aggregate shocks are assumed to be known (Rust, 1994; Magnac and Thesmar, 2002; Arcidiacono and Miller, 2011, 2015). The remaining parameters (and functions) to be identified are: the skill unit production function $\{\tilde{s}_j(h_a)\}_{j\in\{B,W,T\}}$, the variance parameters for wages $\{\sigma_{j\ell}\}_{j\in\{B,W,T\}}^{\ell\in\mathcal{L}}$, the parameters associated to amenities and re-entry costs for the three working alternatives Λ_0, $\{\Lambda_{1j}(\ell)\}_{j\in\{B,W,T\}}$ and $\{\Lambda_{2j}\}_{j\in\{B,W,T\}}$, the parameter associated with the correlation across idiosyncratic shocks ϱ, the deterministic part of the schooling utility function $\tau(\ell, E_a)$, the deterministic component of the home utility $\vartheta(\ell, n_a)$, the equipment capital and STEM parameters in the generation of IPP capital technology χ_1 and χ_2, the parameters of the production function φ, $\tilde{\varsigma}_0$, $\tilde{\varsigma}_1$, $\tilde{\alpha}_0$, $\tilde{\alpha}_1$, $\tilde{\theta}_0$, $\tilde{\theta}_1$, $\tilde{\iota}_0$, $\tilde{\iota}_1$, ρ, κ, and ψ, the parameters from the aggregate shock processes π_ξ, π_ζ, σ_ξ, and σ_ζ, and the capitalist CRRA parameter ν. I discuss the identification of these parameters below building on Lee (1983), Hotz and Miller (1993), Rust (1994), Altuğ and Miller (1998), Magnac and Thesmar (2002), Arcidiacono and Miller (2011, 2015), and Llull (2018a).

The data consist of one-year panels with information on choices, idiosyncratic state variables, and wages for a sample of individuals that is representative of the United States between 1993 and 2023. Additional aggregate data are needed for identification: aggregate output, capital stocks (equipment and structures), stock of ideas (IPP stock), native and immigrant cohort sizes. In practice, I also use data on aggregate wage bills, not needed for identification, and distribution of initials skills (at age 16 for natives, and at entry for immigrants) and of observable types, identified directly from census micro data

A. CCPs

Let $\{p_j(h_a, r_t)\}_{j\in\mathcal{D}(h_a)}$ denote the CCPs. They are not directly identified from the data, as r_t is not observed. In order to recover them, I first note that, given the model specification, calendar time t is a sufficient statistic for r_t^*, the vector of equilibrium skill prices at time t. I also note that the pseudo-CCPs $\{\tilde{p}_j(h_a, t)\}_{j\in\mathcal{D}(h_a)}$ are non-parametrically identified from observed choices by individuals with state vector h_a at time t. As noted below, pseudo-CCPs are sufficient to identify skill prices in the baseline equilibrium, r_t^*. Given identified skill prices, CCPs $\{p_j(h_a, r_t)\}_{j\in\mathcal{D}(h_a)}$ at the recovered skill prices r_t are identified from observed choices by individuals when skill prices are r_t. CCPs at other skill prices are identified by interpolation, assuming there is enough variation in r_t within the sample period. Specifically, I exploit that the only source of non-stationarity in the worker's problem are skill prices, and use $\{p_j(h_a, r)\}_{j\in\mathcal{D}(h_a)}$ as the counterfactual CCPs for an individual with state vector h_a at time t if skill prices were equal to r instead of r_t^*.

B. *Wage function and aggregate skill prices*

The skill production function $\{\tilde{s}_j(h_a)\}_{j\in\{B,W,T\}}$ and the variances of the productivity shock $\{\sigma_{j\ell}\}_{j\in\{B,W,T\}}^{\ell\in\mathcal{L}}$ are identified from individual wage data and the estimated pseudo-CCPs $\{\tilde{p}_j(h_a, t)\}_{j\in\mathcal{D}(h_a)}$. Taking logs to (14) yields:

$$\ln w_j(h_a, \eta_a, r_t) = \ln r_{jt} + \tilde{s}_j(h_a) + \sigma_{j\ell}\eta_a. \tag{36}$$

In the absence of self-selection, skill prices would be identified as time dummies, and $\tilde{s}_j(h_a)$ would be identified as a non-parametric function of h_a. The scale of $\ln r_{jt}$ and $\tilde{s}_j(h_a)$ is not separately identified, so $\tilde{s}_j(h)$ is normalized to zero for one value of the idyosincratic observable state vector h. Given the Roy nature of the labor supply decision, a least squares regression does not identify the skill production function, because $\mathbb{E}[\eta_a|d_{jt} = 1, h_a, t] \neq 0$ (i.e. there is self-selection into working). Alternatively, I follow Lee (1983) to express (36) as:

$$\ln w_j(h_a, \eta_a, r_t) = \ln r_{jt} + \tilde{s}_j(h_a) + \sigma_{j\ell}\omega_{j\ell}\lambda(\tilde{p}_j(h_a, t)) + \tilde{\eta}_{ja}, \tag{37}$$

where $\omega_{j\ell}$ is a nuisance parameter associated to the degree of endogenous self-selection in wages, $\lambda(\tilde{p}_j(h_a, t)) \equiv \phi(\Phi^{-1}(\tilde{p}_j(h_a, t)))/\tilde{p}_j(h_a, t)$ is the selection correction term (where $\phi(\cdot)$, $\Phi(\cdot)$, and $\Phi^{-1}(\cdot)$ are the standard normal density, cumulative distribution function, and its inverse respectively), and $\tilde{\eta}_{ja}$ is the part of the error term that is orthogonal to h_a and t given that the individual decided to work in occupation j. As $\tilde{p}_j(h_a, t)$ is identified, $\lambda(\tilde{p}_j(h_a, t))$ is identified, and, thus, r_t, $\tilde{s}_j(h_a)$, and $\sigma_{j\ell}\omega_{j\ell}$ are identified as the least squares coefficients of (37).

As discussed in the literature (see Vella (1998) for a survey), credible identification requires an exclusion restriction. In this model, the number of children in the household, n_a affects the utility to stay home, but does not affect wages directly, which provides an exclusion restriction. Given this exclusion restriction, the normality assumption could be relaxed, and $\lambda(\tilde{p}_j(h_a, t))$ could still be identified nonparametrically, as noted by Das, Newey and Vella (2003).

Finally, $\sigma_{j\ell}$ is identified from the conditional (residual) variance of wages, which has the following form (Lee, 1983):

$$\mathbb{E}[\tilde{\eta}_a^2|j, h_a, t] = \sigma_{j\ell}^2 - (\sigma_{j\ell}\omega_{j\ell})^2 \left[\Phi^{-1}(\tilde{p}_j(h_a, t)) + \lambda(\tilde{p}_j(h_a, t))\right] \lambda(\tilde{p}_j(h_a, t)). \tag{38}$$

By inspection, $\sigma_{j\ell}$ is identified as all other elements of Equation (38) are identified.

C. *Utility parameters*

Next, we proceed to discuss identification of the parameters associated to amenities and re-entry costs for the three working alternatives Λ_0, $\{\Lambda_{1j}(\ell)\}_{j\in\{B,W,T\}}$ and $\{\Lambda_{2j}\}_{j\in\{B,W,T\}}$, the deterministic parts of the utilities of school and home alternatives, $\{\tau_i(\ell)\}_{i\in\{0,1,2\}}$ and $\vartheta(\ell, n_a)$, and the Generalized Extreme Value parameter ϱ. These parameters are identified using CCP estimation arguments by Hotz and Miller (1993), Hotz et al. (1994), and Arcidiacono and Miller (2011). This approach exploits the mapping between value functions and CCPs to rewrite the dynamic programming problem as a function of the CCPs, data, and parameters. Obtaining non-parametric estimates from the CCPs directly from the data, and replacing them into the CCP representation avoids having to solve the model during estimation.

To proceed with this approach, define the vector of CCPs as $p \equiv (p_B, p_W, p_T, p_S, p_H)'$. Appealing to Hotz and Miller (1993) (with the reformulation in Arcidiacono and Miller, 2011), there exist mappings $\mu_j(p)$ so that:

$$\mu_j[p(h_a, r_t)] \equiv \bar{V}(h_a, r_t) - v_j(h_a, r_t), \text{ for any } j \in \mathcal{D}(h_a). \tag{39}$$

Given the distributional assumption for $F_\varepsilon(\cdot)$ in this model, $\mu_j(p)$ specializes to:

$$\mu_j(p) = \begin{cases} \gamma - \varrho \ln p_j - (1-\varrho)\ln\left(\sum_{k\in\{B,W,T\}} p_k\right) & \text{if } j \in \{B,W,T\} \\ \gamma - \ln p_j & \text{if } j \in \{S,H\}, \end{cases} \tag{40}$$

where γ is the Euler's constant, approximately equal to 0.5772 (see Lemma 3 in Arcidiacono and Miller, 2011). Substituting (39) into (22) yields:

$$v_j(h_a, r_t) = \tilde{u}_j(h_a, r_t) + \beta \int \sum_{h\in\mathcal{H}_{a+1|h_a,j}} [v_k(h,r) + \mu_k(p(h,r))]\, P_h(h|h_a, j) dF_r(r|r_t), \tag{41}$$

for an arbitrary $k \in \mathcal{D}(h_{a+1})$. Since n is the only element of h whose transition probability is not degenerate, $\mathcal{H}_{a+1|h_a,j}$ includes only three elements determined by the three possible values of n. This representation still includes $v_k(h,r)$ and, hence, still requires the solution of the model. One can recursively replace the remaining alternative-specific value functions until none is left. Since (39) holds for any j, there are multiple equivalent representations that can be derived. Given that the likelihood contributions depend on the difference between $v_j(h_a, r_t)$ and a base category $v_1(h_a, r_t)$, a useful representation of each pair of alternative-specific value functions is one in which the remaining continuation values after a certain point are equivalent, which eliminated them upon substraction and avoids having to recursively replace the mappings until the terminal choice. This strategy is defined in Arcidiacono and Miller (2011) as finite dependence. I exploit different finite dependence strategies (different equivalent representations) for the different subsets of parameters to be identified.

To identify work and home utility parameters, I exploit the observation that any sequence of choices in which today's choice is other than school and tomorrow's choice is home leads to the same states two periods ahead with the same probabilities. To see it formally, define $h_j(n, h_a) \equiv (a+1, \ell, E_a, j, n, \tilde{a})'$ as the state vector that follows after a choice j given a specific realization of preschool children n. Given the structure of the model, any pair of choices $(j, k) \in \{B, W, T, H\}^2$ satisfies $h_H(n, h_j(n_a, h_a)) = h_H(n, h_k(n_a, h_a)) = (a+2, \ell, E_a, H, n, \tilde{a})'$. Provided that the fertility process is specified to be invariant to occupation and participation decisions, this implies that if we choose the representation $k = H$ for the term inside the integral in (41), the resulting expression, after substituting (40) in, gives:

$$v_H(h_j, r) + \mu_H(p(h_j, r)) = \vartheta(\ell, n) + \gamma - \ln p_H(h_j, r) + \{\text{cont}\}, \tag{42}$$

where the term $\{\text{cont}\}$ is the continuation value, which does not depend on j and, hence, providing finite dependence one period ahead.

Given the GEV assumption for the errors, conditional choice probabilities have a nested logit form. In the standard nested logit, the limb-specific coefficients within a branch (in this case, occupations within the choice of work) are multinomial logit coefficients scaled by the GEV parameter (in this case, ϱ). In this context, I specify the multinomial logit formulation relative to the base category B, and exploit the representation in (41) for finite dependence. Given common regressors across alternatives, I normalize $\Lambda_{1B}(\ell) = 0$ for all ℓ. Substituting (42) into (41), evaluating the resulting expression for $j \in \{W, T\}$ and for B, and substracting one evaluation by the other gives, upon rearrangement:

$$\begin{aligned} v_j(h_a, r_t) - v_B(h_a, r_t) &= \Lambda_0[\ln r_{jt} + \tilde{s}_j(h_a) - \ln r_{Bt} - \tilde{s}_B(h_a)] + \Lambda_{1j}(\ell) \\ &- (\Lambda_{2j} - \Lambda_{2B}) d_{Ha-1} - \beta \int \sum_{n \in \mathcal{C}} \ln \frac{p_H(h_j(n, h_a), r)}{p_H(h_B(n, h_a), r)} P_n(n|h_a, j) dF_r(r|r_t), \end{aligned} \tag{43}$$

where I exploit that $P_n(n|h_a, j) = P_n(n|h_a, B)$ by assumption, since only education choices affect the fertility process. Provided nonparametric estimates of the CCPs, Λ_0, $\{\Lambda_{1j}(\ell)\}_{j\in\{B,W,T\}}$ and $\{\Lambda_{2j}\}_{j\in\{B,W,T\}}$ are identified as coefficients of the reslting multinomial logit, with the exception of Λ_{2B}, which is identified in the upper level (only $\Lambda_{2j} - \Lambda_{2B}$ are identified in this level). The GEV parameter ϱ is identified both in the upper level, and in this level as the inverse of the estimated coefficient associated to the continuation value, given that the value of β is assumed to be known.

To identify the remaining utility parameters, I exploit the upper level of the nested logit structure. Having identified the parameters from the occupational choice multinomial logit formulation, one can recover the so-called inclusive value or log-sum term for the working alternatives, which I denote by IV. In order to take advantage of the finite dependence argument, it is convenient to express it as:

$$\varrho IV \equiv v_B(h_a, r_t) + \varrho \ln \sum_{j \in \{B,W,T\}} \exp\left[\frac{v_j(h_a, r_t) - v_B(h_a, r_t)}{\varrho}\right]. \tag{44}$$

This formulation facilitates writing the utilities associated to home, school, and work in deviations from the base category $v_B(h_a, r_t)$. For working, this term would correspond to the second element in the right hand side of the above expression. For the home utility, finite dependence is trivially derived following the analogous representation as in (43), namely:

$$\begin{aligned} v_H(h_a, r_t) - v_B(h_a, r_t) = \vartheta(\ell, n_a) - \Lambda_0[\ln r_{Bt} + \tilde{s}_B(h_a)] + \Lambda_{2B} d_{Ha-1} \\ - \beta \int \sum_{n \in \mathcal{C}} \ln \frac{p_H(h_H(n, h_a), r)}{p_H(h_B(n, h_a), r)} P_n(n|h_a, H) dF_r(r|r_t). \end{aligned} \tag{45}$$

Given that d_{Sa-1} does not enter (44) and (45), and that individuals with $d_{Sa-1} = 0$ (the majority of the sample) cannot choose school, $\vartheta(\ell, n_a)$, ϱ, and Λ_{2B} are identified as coefficients of a multinomial logit estimated on the subsample of individuals with $d_{Sa-1} = 0$.

The school representation, on the contrary, is a bit more complicated. In particular, because leaving school is an absorbing state, this decision does not exhibit any form of finite dependence. However, one can still take advantage of the CCP representation to write:

$$\begin{aligned} v_S(h_a, r_t) - v_B(h_a, r_t) &= \Lambda_{2B} d_{Ha-1} - \Lambda_0[\ln r_{Bt} + \tilde{s}_B(h_a)] \\ &+ \tau_0(\ell)\, \mathbb{1}\{E_a < 12\} + \tau_1(\ell)\, \mathbb{1}\{12 \le E_a < 16\} + \tau_2(\ell)\, \mathbb{1}\{E_a \ge 16\} \\ &+ \beta \int \left\{ \begin{array}{l} \sum_{h \in \mathcal{H}_{a+1|h_a,S}} \left[\begin{array}{c} v_H(h, r) \\ - \ln p_H(h, r) \end{array} \right] P_h(h|h_a, S) \\ - \sum_{h \in \mathcal{H}_{a+1|h_a,B}} \left[\begin{array}{c} v_H(h, r) \\ - \ln p_H(h, r) \end{array} \right] P_h(h|h_a, B) \end{array} \right\} dF_r(r|r_t), \end{aligned} \tag{46}$$

and recursively replace $v_H(h, r)$ by its representation in (42) until the terminal period. By inspection, all the elements in the continuation value of that representation are identified, which implies that the elements in $\{\tau_k(\ell)\}_{k\in\{0,1,2\}}$ are also identified as the remaining unrestricted coefficients of a multinomial logit on the subsample of individuals with $d_{Sa-1} = 1$.

D. Production function

I now proceed with the the key equations and restrictions that identify the production function parameters, with the exception of φ and the aggregate shock processes, which are discussed in the next sub-section. In particular, I show that one can recursively identify these parameters from a reduced number of aggregate linear regressions on the available time-series of aggregate data and recovered aggregate skill units and skill prices, identified from the arguments above. In practice, however, the short time-series available for identified labor market aggregates (recall that the micro-data covers the period 1993-2023) impedes a precise-enough estimation of the model parameters using this method. For this reason, I estimate these parameters and those discussed in the next sub-section by Generalized Method of Moments (GMM) imposing, together, all the implied moment conditions associated to the equations I describe in this section, plus a set of additional restrictions implied by the equilibrium conditions of the model, as discussed in Section V.D below.

Given identified aggregate skill units and their returns (or, in practice, given data on aggregate wage bills), the capital share Γ_K is identified. Capital stocks K_{St} and $q_{Et}K_{Et}$ are also observed in the data (given that separate data on dollar value of capital equipment and relative prices of equipment are available) and, hence, total assets are identified as the sum of the two. Imposing that the intermediary's budget constraint (32) holds with equality in equilibrium, the implicit return to assets is given by:

$$r_{At} = \frac{\Gamma_K Y_t + (1-\delta_{St})K_{St} + (1-\delta_{Et})q_{Et}K_{Et}}{K_{St} + q_{Et-1}K_{Et}} - 1, \tag{47}$$

where all the elements in the right hand side are observable in the data. Given the no-arbitrage constraint, r_{Et} and r_{St} are also identified from (34).[25]

Aggregating identified individual skill units over all individuals employed in each occupation identifies equilibrium aggregate skill units. Rewriting Equation (10)

[25] In equilibrium, these returns are also identified from the equations below. Therefore, the no-arbitrage condition provides an overidentifying restriction. In the estimation of production function parameters I take capital stocks from the data and hence use the no-arbitrage equation as a moment condition in my criterion function. When I solve for capital stocks in equilibrium, on the contrary, I impose no-arbitrage.

as a factor share Γ_{St}, deriving the analogous expression for $1-\Gamma_{St}$, dividing the former by the latter, and taking logs yields:

$$\ln \frac{\Gamma_{St}}{1-\Gamma_{St}} = \ln \frac{\varsigma_t}{1-\varsigma_t} = \tilde{\varsigma}_0 + \tilde{\varsigma}_1 \frac{I_t}{Y_t}. \tag{48}$$

Thus, given structures shares Γ_{St} and ideas to output ratios I_t/Y_t can be obtained from the data and the identified interest rates, the parameters $\tilde{\varsigma}_0$ and $\tilde{\varsigma}_1$ are identified in the above expression as regression coefficients of log relative shares on a constant and the ideas-to-output ratio. Combining Equations (7) and (11) and taking logs gives, upon rearrangement:

$$\ln \frac{r_{Tt}}{r_{Et}} = \tilde{\iota}_0 + \tilde{\iota}_1 \frac{I_t}{Y_t} + (\psi - 1) \ln \left(\frac{S_{Tt}}{K_{Et}} \right). \tag{49}$$

Equation (49) provides the basis for identification of $\tilde{\iota}_0$, $\tilde{\iota}_1$, and ψ, which can be obtained from regression coefficients. Having identified these parameters, Q_{1t} is identified. Similarly, combining Equations (7) and (8) and taking logs to the resulting expression and rearranging yields:

$$\ln \frac{r_{Wt}}{r_{Tt}} + \ln \iota_t + (\psi - 1) \ln \frac{S_{Tt}}{Q_{1t}} = \tilde{\theta}_0 + \tilde{\theta}_1 \frac{I_t}{Y_t} + (\kappa - 1) \ln \left(\frac{S_{Wt}}{Q_{1t}} \right), \tag{50}$$

which identifies $\tilde{\theta}_0$, $\tilde{\theta}_1$, and κ from regression coefficients. Proceeding analogously with Equations (8) and (9), $\tilde{\alpha}_0$, $\tilde{\alpha}_1$, and ρ are also identified from:

$$\ln \frac{r_{Bt}}{r_{Wt}} + \ln \theta_t + (\kappa - 1) \ln \frac{S_{Wt}}{Q_{2t}} = \tilde{\alpha}_0 + \tilde{\alpha}_1 \frac{I_t}{Y_t} + (\rho - 1) \ln \left(\frac{S_{Bt}}{Q_{2t}} \right). \tag{51}$$

E. Capital supply, ideas, and shocks

The remaining parameter of the production function, φ, those of the production function of ideas, χ_1, and χ_2, and those associated to the aggregate shock processes π_ξ, π_ζ, σ_ξ, and σ_ζ are identified as follows. Having identified ς_t, α_t, ρ, θ_t, κ, ι_t, and ψ, the term $\zeta_t I_t^\varphi$ is identified as the residual in Equation (3), which I denote by z_t. Taking logs and first differences, and substituting Equation (5) into the resulting expression yields:

$$\Delta \ln z_t = \pi_\zeta + \varphi \Delta \ln I_t + \sigma_\zeta \upsilon_{\zeta t}. \tag{52}$$

Even though $\Delta \ln I_t$ is correlated with $\upsilon_{\zeta t}$ because $\ln I_t$ is, $\ln I_{t-1}$ (and any other lag) is a valid instrument because it is correlated with $\Delta \ln I_t$ but not with $\upsilon_{\zeta t}$. Thus, π_ζ and φ are identified as (instrumental variable) regression coefficients, and σ_ζ is identified as the variance of the residual. Similarly, taking logs and first differences to (1), and substituting Equation (2) into the resulting expression, we obtain, upon rearrangement:

$$\Delta \ln(\Delta I_t) = \pi_\xi + \chi_1 \Delta \ln K_{Et} + \chi_2 \ln S_{Tt} + \sigma_\xi \upsilon_{\xi t}. \tag{53}$$

Thus, π_ξ, χ_1, χ_2, and σ_ξ are identified in an analogous way using $\ln K_{Et-1}$ and $\ln S_{Tt-1}$ as an instrument for $\Delta \ln K_{Et}$ and $\Delta \ln S_{Tt}$. Finally, the capitalists CRRA parameter ν is identified from the observed assets transitions and interest given the implied policy function $\mathfrak{s}(r_{At}; \beta, \nu)$ taking the parameters for the expectation rule for assets obtained directly from implied assets returns from the data.

V. Estimation

I proceed with estimation following a stepwise procedure that builds on the identification arguments above. To do so, I combine aggregate data from different sources with micro data from the March Supplements of the CPS linked over two consecutive years for the period 1993–2023. First, I estimate the CCPs. Second, I estimate the parameters of the wage equation. Third, I proceed with the estimation of the utility parameters. And finally, I estimate the representative firm problem. Variable definitions and sample selection are specified in Appendix A.

A. CCPs and transition functions

The nonparametric estimates of the preudo-CCPs, $\hat{\hat{p}}(h_a, t)$, are obtained from running flexibly specified multinomial logit models on different subsamples. Because previous choice and having a college degree determine the choice set, these two characteristics define the subsamples. For the two sub-samples whose previous choice is not school, I estimate a multinomial logit with a specification of the regressors that closely mimics the flow utilities, in a more flexible specified way (e.g., the elements of the wage function enter individually instead of comined in the form of a predicted wage, or I do not group types as much as I do in the flow utility). To each of the specifications, I add a full set of time dummies to each of the working utilities, to generate the sufficient statistics of skill prices in the pseudo-ccps. The probability of attending school is set to zero, $\hat{p}_S = \hat{\hat{p}}_S = 0$, when $d_{a-1} \neq S$, and the probability of working in STEM is set to zero, $\hat{p}_T = \hat{\hat{p}}_T = 0$, if the worker does not possess a college degree, $E < 16$. Having estimated these first two sets of multinomial logits, the probabilities of attending school are estimated in the remaining two sub-samples using a binary logit (attending school versus not) including the elements of the school utility as regressors and adding the predicted probabilities of working in each alternative obtained using the estimates of the first two subsamples. For these subsamples, the probabilities are then predicted as the product of the predicted probabilities of working in each alternative conditional on not going to school multiplied by the probability of attending school.

After estimating the wage equation (see below), I recover estimates of time dummies in each regression, which are estimates of the baseline equilibrium (log) skill prices $\ln \hat{r}_{jt}$. CCPs are then re-estimated replacing time dummies with a quadratic polynomial in $\ln \hat{r}_{jt}$.

The transition functions are all degenerate except for $P_n(n|h_a, d_a)$. As noted above, this function is assumed to depend on the current choice only if this choice is schooling (and only if $E_a = 15$, in which case the choice of schooling gives the individual a college degree in the next period). Furthermore, I assume that the dependence on h_a is through type, education level, age, and current number of children. The transition probability matrix is estimated nonparametrically using Census data for 1970–2000, and ACS data for 2001–2023. The probabilities are estimated on subsamples determined by type, education level, and current number of children. The dependence on age is estimated using a logit specification in a flexible polinomial in age.

B. *Wage function*

Even though $\tilde{s}_j(h_a)$ is nonparametrically identified, I parametrize it to obtain more precise estiamtes. Let $X_a \equiv a - \max\{16, E+6\}$ denote potential experience, and $\tilde{X}_a \equiv \max\{0, \tilde{a} - E_a - 6\}$ denote potential experience abroad. The function $\tilde{s}_j(h_a)$ specializes to the following Mincerian regression (Mincer, 1974):

$$\begin{aligned}\tilde{s}_j(h_a) \equiv {} & (\flat_{1j} + \flat_{2j}\, \mathbb{1}\{\tilde{\ell}(\ell) \in \tilde{\mathcal{L}}_3 \cup \tilde{\mathcal{L}}_4\} + \flat_{3j}\, \mathbb{1}\{d_{a-1} \neq j\}) \times E_a \\ & + \flat_{4j}\, \mathbb{1}\{E_a = 12\} + \flat_{5j}\, \mathbb{1}\{E_a \in [13, 15]\} + \flat_{6j}\, \mathbb{1}\{E_a \geq 16\} \\ & + (\flat_{7j} + \flat_{8j}\, \mathbb{1}\{d_{a-1} \neq j\}) \times X_a \\ & + (\flat_{9j} + \flat_{10j}\, \mathbb{1}\{d_{a-1} \neq j\}) \times X_a^2 \\ & + \flat_{11j} \tilde{X}_a + \flat_{12j}\, \mathbb{1}\{d_{a-1} \neq j\} + \sum_{k \in \mathcal{L}} \flat_{13kj}\, \mathbb{1}\{k = \ell\},\end{aligned} \tag{54}$$

where $\tilde{\ell}(\ell) \in \tilde{\mathcal{L}}_3 \cup \tilde{\mathcal{L}}_4$ denotes that the type ℓ is included in the immigrant subset. Substituting this expression into Equation (37), and evaluating $\lambda(p)$ at the CCPs $\tilde{p}_j(h_a, t)$ estimated in the previous step, the wage function parameters are estimated by least squares estimates on the resulting expression. The equilibrium skill prices are obtained as the coefficients associated to calendar time dummies.

This procedure provides estimates for $\widehat{\ln r_{jt}}$, $\hat{\hat{s}}_j(h)$, and $\widehat{\sigma_{j\ell}\omega_{\ell j}}$. The variance parameter $\sigma_{j\ell}$ is obtained from the sample analogs of moment conditions implied by (38). In particular, by the law of iterated expectations, (38) implies:

$$\sigma_{j\ell}^2 = \mathbb{E}\left[\tilde{\eta}_a^2 + (\sigma_{j\ell}\omega_{j\ell})^2 \left[\Phi^{-1}(\tilde{p}_j(h_a,t)) + \lambda(\tilde{p}_j(h_a,t))\right] \lambda(\tilde{p}_j(h_a,t)) \middle| d_{ja} = 1, \ell\right]. \tag{55}$$

A consistent estimator is provided by the sample analog of this expression using the estimated CCPs and $\widehat{\sigma_{j\ell}\omega_{\ell j}}$. To obtain more precise estimates, I only allow $\sigma_{j\ell}$ to differ by sex and nativity status (native/immigrant), this is, I assume $\sigma_{j\ell} = \sigma_{j\ell'}$ if $\tilde{\ell}(\ell) = \tilde{\ell}(\ell')$.

C. *Utility parameters*

The estimation of the utility parameters builds on the nested logit CCP representation outlined in the previous section. The different functions are parameterized as follows. The value of consumption relative to amenity Λ_0 is common across individuals. The amenity value of each occupation $\{\Lambda_{1j}(\ell)\}_{j\in\{T,W\}}$ is parameterized as sex/nativity status dummies. The occupation-specific reentry cost $\{\Lambda_{2j}\}_{j\in\{B,W,T\}}$ is common across different individuals but differs across occupations. Home utility $\vartheta(\ell, n_a)$ is parameterized as:

$$\vartheta(\ell, n) \equiv \vartheta_{0\ell} + \sum_{k=1}^{4} \vartheta_{1k}\, \mathbb{1}\{\tilde{\ell}(\ell) \in \tilde{\mathcal{L}}_k\} n. \tag{56}$$

In words, $\vartheta_{0\ell}$ denotes the type-specific intercept, and ϑ_{1k} for $k \in \{1,2,3,4\}$ denote how this utility is shifted by each children in the household respectively for native male, native female, immigrant male, and immigrant female. Finally, school utilities are parameterized as:

$$\tau_0(\ell) \equiv \tau_{0\ell}; \quad \tau_1(\ell) \equiv \tau_{0\ell} + \tau_1; \quad \tau_2(\ell) \equiv \tau_{0\ell} + \tau_2, \tag{57}$$

or, in words, a type-specific school utility $\tau_{0\ell}$ shifted by common additional tuitions for college and graduate school τ_1 and τ_2. Similar to the working alternatives, $\tau_{0\ell}$ is parameterized as a set of sex/nativity status dummies.

Before implementing the nested logit estimation, the terms associated to the integrals in (43), (45) and (46) need to be computed. Given that the assumed expectation rule for skill prices in (25), skill price innovations are over-time independent and identically distributed random normal innovations correlated across occupations. To forecast future wages, I thus draw from a trivariate normal distribution with means Ξ_0 and covariance structure $\Sigma_{\ln r}$ directly estimated from the recovered skill prices in the wage estimation.

For each individual, the continuation value term for each $j \in \{T, W, H\}$ is obtained as:

$$\hat{\Omega}_j^{(i)} \equiv 0.95 \int \sum_{n \in \mathcal{C}} \frac{\hat{p}_H \left(h_j(n, h^{(i)}), r(\upsilon_{\ln r_{t+1}}, \hat{r}_t)\right)}{\hat{p}_H \left(h_B(n, h^{(i)}), r(\upsilon_{\ln r_{t+1}}, \hat{r}_t)\right)} \hat{P}_n(n|h^{(i)}, j) d\Phi_3(\upsilon_{\ln r_{t+1}}; \Xi_0, \Sigma_{\ln r}), \tag{58}$$

where $\upsilon_{\ln r_{t+1}}$ is the predicted vector innovations in skill prices implied by Equation (25). This integral is computed using a three-dimensional Gauss-Chebychev quadrature with seven points per dimension.

The school-vs-blue collar continuation value in (46) is more costly computationally. For this reason, I follow Hotz et al. (1994) and Altuğ and Miller (1998) and use simulation methods to approximate it. In particular, for each individual i, I simulate $M(= 80)$ sequences of skill prices and children (denoted by m_i) for periods $l_i \in \{1, ..., 65 - a_i\}$, drawing from the skill prices expectation rule and the estimated children transition functions, as for the work or home continuation value, at each simulation point (m_i, l_i). Thus, in each simulation m_i, the individual is exposed to a sequence of skill prices $\{r_{t_i+l_i}^{(m_i)}\}_{l_i=1,...,65-a_i}$, and a five sequences of state variables, depending on the initial choice: $h_{jl_i}(n_{jl_i}^{(m_i)}, h^{(i)}) \equiv (a_i+l_i, \ell_i, E^{(i)}+\mathbb{1}\{j = S\}, j, n_{jl_i}^{(m_i)}, \tilde{a}_i)$. The difference across sequences is as follows: at $l_i = 1$, the previous choice varies across paths (school in the path associated to $j = S$, and blue collar in the path associated to $j = B$); education increases in one unit if $j = S$ and stays constant otherwise; and there are two sequences of children draws, depending on whether the choice is school or not (the distinction between $n_{Sl_i}^{(m_i)}$ and $n_{jl_i}^{(m_i)}$ is necessary because children transition probabilities vary by education, but $n_{jl_i}^{(m_i)} = n_{j'l_i}^{(m_i)}$ for any $j, j' \neq S$ for the same reason). Using these simulations, the last term of (46) for individual (i) is approximated by:

$$\hat{\Omega}_S^{(i)} = \frac{1}{M} \sum_{m_i=1}^{M} \sum_{l_i=1}^{65-a_i} 0.95^{l_i} \left[\begin{array}{r} \displaystyle\sum_{k=1}^{4} \hat{\vartheta}_{1k} \, \mathbb{1}\{\tilde{\ell}(\ell_i) \in \tilde{\mathcal{L}}_k\} \left(n_{Sl_i}^{(m_i)} - n_{Bl_i}^{(m_i)}\right) \\ \displaystyle - \ln \frac{\hat{p}_H \left(h_{Sl_i}(n_{Sl_i}^{(m_i)}, h^{(i)}), r_{t_i+l_i}^{(m_i)}\right)}{\hat{p}_H \left(h_{Bl_i}(n_{Bl_i}^{(m_i)}, h^{(i)}), r_{t_i+l_i}^{(m_i)}\right)} \end{array} \right]. \tag{59}$$

Using these constructs, the estimation of the utility parameters reduces to a (multi-step) nested logit procedure. In practice, the estimation is carried out as follows. The parameters ϱ and Λ_0 enter at different stages and are therefore obtained in an outer loop that jointly maximizes the likelihoods of the inner steps using a Nelder-Mead algorithm. In the inner loop, the parameters of white-collar and STEM utilities are estimated as coefficients of a multinomial logit based on (43) (with all coefficients divided by ϱ, as discussed in the text), replacing the last term with (58). This step is estimated on the subsample of individuals who choose to work (and whose previous choice was not school), taking ϱ and Λ_0 as fixed at their current values in the outer loop. Next, the parameters of the home alternative are estimated in a second step, again holding ϱ and Λ_0 fixed, using a logit specification based on (44) and (45) for the decision to work versus not work. The inclusive value (44) is computed using the parameter estimates from the first stage. This second step uses also individuals who decide to not work, but still excludes those whose previous-period choice was school. Finally, a third step estimates the school utility parameters via a multinomial logit with work, school, and home as alternatives, based on (44), (45), and (46), using the subsample of individuals whose previous-period choice was school.

To increase the precision of the estimates, and increase model's internal consistency, in I proceed with an iterative algorithm in the spirit of Aguirregabiria and Mira (2002). After the initial estimation of the parameters, I solve the value functions at the estimated parameters. This solution gives a consistent update of the continuation values and the CCPs. Using these predictions, I obtain a new set of consistent estimates using the algorithm described above. As in Aguirregabiria and Mira (2002), iteration until convergence is asymptotically equivalent to a full solution approach, but it is not a requirement, since the estimates obtained in each iteration are consistent. I implement this iterative procedure until convergence before running the production function estimation. However, given the computationally more demanding nature of the estimation step for ϱ and Λ_0, I do not update these two parameters when I iterate. Given that the first set of estimation results is already consistent, these parameter estimates remain consistent.

D. Production function parameters

In theory, the parameters of the production function can be estimated sequentially by least squares using aggregate skill units constructed from the estimated individual skill units production function (and skill prices). In practice, however, these aggregate skill units would only be computed since 1993, which leaves a limited time series to precisely estimate these parameters. Alternatively, my estimation procedure imposes all the restrictions described in Section IV.D plus all the implied equilibrium restrictions that result from solving the model at the estimated utility parameters. To do that, I formulate a Generalized Method of

Moments (GMM) estimation with the orthogonality conditions that follows. I then minimize the squares of these moment conditions (weighted by the identity matrix) using the BOBYQA algorithm for bound constraint optimization without derivatives (Powell, 2009). To do so, I solve the model as described in Section V.F below. My GMM estimation uses 41 moment conditions. A first set of moment conditions is based on Equations (48) to (51). Defining four residuals (the fifth is collinear because factor shares sum to one) as the difference between the factor shares in the data and the implied factor shares in the model, the first set of moment conditions orthogonalizes these residuals with respect to a constant, the ideas to output ratio I_t/Y_t, and the corresponding input ratios that appear, respectively, in each of the equations. In a second set of moment conditions, I compare some aggregate moments obtained from the CPS and their simulated counterparts. I only orthogonalize these differences with respect to a constant. All these moments are separately computed for men and women. These moments include: average years of education, participation rate, share of workers in STEM and in white collar, average log annual wages, college to high school wage gap, and STEM to blue collar and white collar to blue collar wage gaps. A third set of moment conditions are based on a zero expected value for the no-arbitrage condition (not imposed in this version of the solution as described in Footnote 25). Finally, I also include a fourth set of moment conditions based on the expected difference between the recovered skill prices and their counterparts in the wage estimation (note that this moment condition is only available for a shorter period of time). Using these procedure I estimate the 11 parameters described in Section IV.D.[26]

E. Capital supply, ideas, and shocks

After estimating the production function parameters using capital stocks from the data, I then proceed to the estimation of the remaining parameters of the model (φ, the parameters of the shock processes, the CRRA parameter ν, and the baseline expectation rule for capital). To do that, I proceed in a similar way than with the production function parameters. Specifically, I now work with a full solution of the equilibrium that also solves for the capital supply instead of taking capital stocks from the data. I proceed with a GMM estimation implemented using the BOBYQA algorithm as described above. I use a total of 41 moment conditions. A first set of moment conditions is given by the log of assets, structures, and equipment capital stocks. A second set is given by the same moments used above

[26] This estimation sometimes gets stuck in local minima. The estimates reported in the paper are obtained starting from different starting values and selecting the global minimum achieved.

from the CPS. A third set of moments are based on the orthogonality conditions described in Equations (52) and (53). For those, I use contemporaneous assets (assets in period t are predetermined and, hence, do not depend on $\upsilon_{\zeta t}$ or $\upsilon_{\xi t}$) and lagged capital stocks and aggregate skill units (only assets, equipment and STEM skill units for υ_{ξ}, and all the inputs plus lagged stock of ideas for υ_{ζ}). Finally, a last set of moment conditions are mean and variance moment conditions for the aggregate shock processes.[27]

F. Model solution

The model solution begins with the value functions. The value functions are solved for all ages, for all demographic types, 0, 1, and 2+ children, all previous period choices, and random subsets of combinations of ages of arrival (or potential experience abroad), years of education, and skill prices. Solution is by backwards induction. For every age, after solving the value function at all the evaluation points, an interpolation regression is estimated to determine the value at the age of arrival, years of education, and skill price combinations in which the function is not evaluated. A separate interpolation regression is estimated for each age, type, children, previous period choices, and college/non college. Each of this regressions includes a rich combination of polynomials and pairwise interactions of the five variables to be interpolated, and is estimated on 10,000 random draws of the possible combinations of these variables. The solution algorithm is implemented in each iteration of the Aguirregabiria and Mira (2002) updates of the utility estimation, once before the estimation of the production function and capital supply parameters, and once before each counterfactual scenario.

The second step of the solution algorithm is the simulation of the economy. The equilibrium is obtained by simulating the behavior of successive cohorts of 50,000 individuals from 1860 to 2023. Since the production function is estimated from moment conditions that start in 1967, two entire generations have initalized the model before that date. From 1860 to 1900, the model is initalized with aggregate conditions of 1900. Starting that year, aggregate conditions start evolving as in the data. For each individual, choices are simulated drawing idiosyncratic shocks following the procedure outlined in Appendix B.

Three different algorithms are implemented to solve equilibrium skill prices at different stages of the estimation and simulations. The simplest one is imple-

[27] Similar to the estimation of production function parameters, this estimation sometimes gets stuck in local minima. The estimates reported in the paper are obtained starting from different starting values and selecting the global minimum achieved.

mented in the estimation of the production function parameters. Equilibrium skill prices are recovered as a sequence of fixed points. For an initial guess of skill prices, I obtain the supply of skills in each occupation. Taking the stock of ideas I_t, output Y_t, capital stocks K_{Et} and K_{St}, and returns to assets r_{At} as given, I plug this initial supply of skills in the firm's first order condition to obtain an update of the skill prices and returns to equipment (returns to structures are completely determined by output and structures). I iterate this procedure until I reach a fixed point for each period. In this version of the algorithm, the no-arbitrage condition is not imposed in equilibrium, and, hence, it is used as a moment condition. The second algorithm, used in the estimation of the capital supply, ideas, and shocks parameters, internalizes the capital supply. In an initial step I solve for the policy function that determines the savings rate and, hence, the capital supply function. Then the economy is simulated as above, but updating also assets and capital stocks based on the capital supply and the no-arbitrage condition. In this procedure only output Y_t and the stock of ideas I_t are taken from the data in order to identify aggregate shocks. Finally, the third algorithm, used for counterfactual simulations, uses the baseline aggregate shocks instead of using I_t and Y_t.

A final step of the solution algorithm is solving rational expectations. In the evaluations during estimation and baseline simulations, the expectation rule for skill prices is taken from the data, and hence I do not implement an updating fixed point to obtain expectation rules that coincide with the recovered sequences of skill prices. In counterfactual simulations, an outer loop updates the expectation rules so that the parameters used for expectations coincide to those obtained by fitting a regression on the counterfactual evolution of skill prices and asset returns.

VI. Parameter Estimates and Goodness of Fit

This section presents parameter estimates and discusses the ability of the model in fitting different patterns from the data. The section is structured as follows. First, I discuss the labor supply estimates and evaluate them with predictions from a partial equilibrium version of the model in which I take the estimated skill prices from the wage regressions as parameters. Then, I go over production function parameter estimates, which are obtained taking the stocks of capital and ideas directly from the data. In combination, I also evaluate the goodness of fit of this equilibrium version of the model. Third, I discuss the remaining parameter estimates (capitalists, ideas, and shocks), and I evaluate the goodness of fit of the model in the full solution equilibrium with non-updated expectation rules. I then obtain equilibrium values for the expectation rule parameters, I discuss the role of expectations in the model, and I evaluate the goodness of the full solution equilibrium in fitting the data. And finally, I evaluate the ability of the model in fitting immigrant choices and wages in and out of sample.

A. *Labor supply parameters and partial equilibrium fit*

Wage function estimates are presented in Table 2. Estimated parameters suggest an important role for education, especially in white collar and STEM occupations. Immigrants experience a lower return to education, consistent with the absence of perfect portability of education across countries. Potential experience shows the standard convex profile, and returns to potential experience abroad are lower than those of domestic experience suggesting downgrading at arrival and generating a form of immigrant wage assimilation as the bundle of experience becomes prominently domestic. Reentry costs in a given occupation are increasing (in absolute value) in education and potential experience. For STEM, these costs are above 0.1 log points in essentially all feasible education levels, and they are almost flat in the first 20 years of potential experience but substantially increasing afterwards. For white collar they are approximately flat, and range from nearly zero to 0.15 log points depending on education. For blue collar, patterns are also relatively flat and increasing in education (they are even positive for very low educational levels). There are significant differences in wages between male and female, sustaining gender wage gaps in equilibrium. Immigrant intercepts are very positive in STEM occupations (suggesting positive selection for highly educated immigrants) and more moderate for white collar and blue collar occupations, being negative for male only for Mexican workers in blue and white collar, and Central Americans and Other Asia/Africa in white collar only. For natives, wage gaps of Hispanics and Blacks are negative and large. Finally, wage variances are relatively large, as a result of being identified from cross-sectional variation. The common productivity shock for all occupations and the labor market reentry cost avoid excessive job transitions as shown below.

Table 3 reports the utility parameters. The estimates imply that labor-supply choices respond strongly to pecuniary incentives, as shown by the estimated consumption utility parameter Λ_0. However, the estimates also point to substantial non-pecuniary components in occupational choices. The work re-entry costs are very large and very similar across occupations, indicating that transitions back into employment after a period at home are costly regardless of the occupation entered. Conditional on working, the occupation-specific amenities show strong differences across demographic groups and occupations. Relative to blue col-

Table 2—Wage Function Parameters

	STEM		White collar		Blue collar	
Education ($\flat_{1j}$)	0.063	(0.002)	0.074	(0.002)	0.041	(0.003)
Education×immigrant ($\flat_{2j}$)	-0.019	(0.005)	-0.013	(0.002)	-0.022	(0.002)
Education×$d_{a-1} \neq j$ ($\flat_{3j}$)	-0.007	(0.006)	-0.007	(0.002)	-0.008	(0.002)
High school graduate ($\flat_{4j}$)			0.027	(0.010)	0.087	(0.007)
Some college ($\flat_{5j}$)			0.030	(0.012)	0.146	(0.012)
College graduate ($\flat_{6j}$)			0.174	(0.016)	0.303	(0.017)
Potential experience ($\flat_{7j}$)	0.035	(0.001)	0.033	(0.001)	0.031	(0.001)
Potential exp.×$d_{a-1} \neq j$ ($\flat_{8j}$)	0.002	(0.003)	-0.001	(0.001)	-0.004	(0.001)
Potential exp. sq. ($\flat_{9j} \times 100$)	-0.059	(0.003)	-0.054	(0.001)	-0.049	(0.001)
$X_a^2 \times d_{a-1} \neq j(\flat_{10j} \times 100)$	-0.004	(0.007)	0.001	(0.002)	0.007	(0.002)
Potential exp. abroad ($\flat_{11j}$)	-0.009	(0.001)	-0.009	(0.001)	-0.008	(0.000)
Previous choice$\neq j$ ($\flat_{12j}$)	0.006	(0.100)	0.019	(0.028)	0.097	(0.028)
Female intercepts ($\flat_{13\ell j}$):						
Western countries	-0.022	(0.114)	-0.179	(0.040)	-0.179	(0.037)
Mexico	-0.138	(0.127)	-0.293	(0.039)	-0.295	(0.030)
Central America & Carib.	-0.093	(0.117)	-0.270	(0.039)	-0.284	(0.032)
South America	-0.046	(0.116)	-0.274	(0.044)	-0.260	(0.040)
Selected Southeast Asia	0.042	(0.111)	-0.176	(0.042)	-0.206	(0.037)
India	0.177	(0.105)	-0.136	(0.053)	-0.154	(0.068)
Other Asia and Africa	0.017	(0.112)	-0.211	(0.042)	-0.216	(0.034)
Native Hispanic	-0.408	(0.069)	-0.407	(0.012)	-0.424	(0.026)
Native Non-Hispanic Black	-0.433	(0.066)	-0.451	(0.011)	-0.460	(0.023)
Native Non-Hisp.Non-Black	-0.379	(0.065)	-0.379	(0.010)	-0.431	(0.023)
Male intercepts ($\flat_{13\ell j}$):						
Western countries	0.542	(0.108)	0.221	(0.050)	0.183	(0.031)
Mexico	0.299	(0.114)	-0.037	(0.049)	-0.067	(0.026)
Central America & Carib.	0.346	(0.105)	-0.026	(0.053)	-0.020	(0.028)
South America	0.422	(0.106)	0.036	(0.058)	0.019	(0.033)
Selected Southeast Asia	0.437	(0.102)	0.049	(0.053)	0.025	(0.034)
India	0.581	(0.097)	0.198	(0.059)	0.144	(0.055)
Other Asia and Africa	0.400	(0.099)	-0.014	(0.056)	0.001	(0.031)
Native Hispanic	-0.061	(0.026)	-0.087	(0.010)	-0.072	(0.008)
Native Non-Hispanic Black	-0.093	(0.024)	-0.144	(0.009)	-0.136	(0.007)
Native Non-Hisp.Non-Black	0		0		0	
Variances of wages ($\sigma_{\tilde{\ell}j}$):						
Female Immigrant	0.273	(0.013)	0.317	(0.006)	0.253	(0.005)
Female Native	0.244	(0.005)	0.278	(0.002)	0.326	(0.004)
Male Immigrant	0.341	(0.009)	0.419	(0.009)	0.279	(0.004)
Male Native	0.297	(0.005)	0.385	(0.003)	0.314	(0.002)

Note: Aguirregabiria and Mira (2002) estimation. Standard errors, in parentheses, are regression-based, obtained in the last iteration of the Aguirregabiria-Mira algorithm.

lar work, the omitted occupation in the amenity normalization, both STEM and white collar work carry negative non-pecuniary terms for all groups, with substan-

Table 3—Work Utility Parameters

	Work					
	STEM		White collar		Blue collar	
Re-entry cost (Λ_{1j})	-3.335	(0.012)	-3.283	(0.010)	-3.245	(0.010)
Amenities ($\Lambda_{2\ell j}$):						
Female Immigrant	-0.942	(0.002)	-0.194	(0.001)	0	
Female Native	-0.585	(0.001)	0.093	(0.001)	0	
Male Immigrant	-1.347	(0.002)	-0.422	(0.001)	0	
Male Native	-0.707	(0.001)	-0.112	(0.001)	0	
Consumption utility (Λ_0)	1.722	(0.000)				
GEV parameter (ϱ)	0.211	(0.000)				

School		
Baseline amenities ($\tau_{0\ell}$):		
Female Immigrant	19.202	(0.043)
Female Native	19.017	(0.023)
Male Immigrant	18.483	(0.043)
Male Native	18.798	(0.019)
College tuition (τ_1)	-3.076	(0.039)
Graduate tuition (τ_2)	-1.877	(0.092)

Home				
Baseline amenities ($\vartheta_{0\ell}$):	Female		Male	
Western countries	18.067	(0.040)	18.287	(0.052)
Mexico	17.617	(0.035)	17.427	(0.045)
Central America & Caribbean	17.621	(0.045)	17.648	(0.058)
South America	17.791	(0.074)	17.777	(0.100)
Selected Southeast Asia	18.103	(0.042)	18.061	(0.062)
India	18.504	(0.076)	18.093	(0.108)
Other Asia and Africa	17.965	(0.046)	17.886	(0.052)
Native Hispanic	17.982	(0.031)	18.284	(0.039)
Native Non-Hispanic Black	18.019	(0.023)	18.396	(0.026)
Native Non-Hispanic Non-Black	18.260	(0.009)	18.554	(0.010)
Children amenities ($\vartheta_{1\tilde{\ell}}$):				
Immigrant	0.282	(0.036)	-0.091	(0.050)
Native	0.136	(0.016)	-0.342	(0.024)

Note: Aguirregabiria and Mira (2002) estimation. Standard errors, in parentheses, are regression-based, obtained in the last iteration of the Aguirregabiria-Mira algorithm. Standard errors for Λ_0 and ϱ are carried over from the one-step Nelder-Mead profile-likelihood Hessian.

tially larger disamenities in STEM. These disamenities, are larger for immigrants and for men, reconcile the substantial wage gaps and the observed differences in participation in each occupation. The low estimated GEV parameter indicates strong correlation in the unobserved components of the three work alternatives, which is intuitive given that the three work options share common features , including the common productivity shock. Together, these estimates imply that workers do not reallocate across occupations as if jobs were frictionless substitutes: changes in skill prices induced by immigration translate into occupational adjustments only gradually and heterogeneously across groups.

The school and home utility estimates discipline the education and participation margins of the model. Baseline school amenities are large for all four broad demographic groups, while the negative college and graduate tuition parameters capture the additional utility cost of remaining in school at higher education levels. The home-utility estimates reveal substantial heterogeneity by sex, nativity, and origin group. Baseline home amenities are generally higher for men than for women within the same origin group, while the effect of preschool children differs sharply by sex: children raise the value of home time for women, especially immigrant women, but reduce it for men, with a particularly large negative effect for native men. This pattern is consistent with observed adjustments in labor market participation after child bearing, which is typically reduced by women and marginally increased by men.

VII. Counterfactual Simulations and Policy Analysis

VIII. Conclusions

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Appendix A: Sample Selection and Variable Definitions

The model is estimated using micro data from the Current Population Survey (CPS). Additionally, some aggregate macro data are used in the estimation and solution of the model, as described in the main text. While the estimation period for the labor supply is 1993–2023, and for the labor demand is 1967–2023, in my simulations I initialize the model starting in 1860 in order to eliminate the influence of initial conditions. In particular, I simulate the first 40 years, i.e. the period 1860–1899, using the aggregates of 1900. Then I simulate the remaining years with actual macro data. As a result, two entire generations go by before the first year of estimation. This appendix provides a detailed description of the main data sources and the main data cleaning procedures.

A1. Aggregate data

Output. Output is measured as Real Gross Domestic Product at chained U.S. dollars of 2017. The raw data is provided by the Bureau of Economic Analysis (BEA) in the National Income and Product Accounts (NIPA), Table 1.1.6 "Real Gross Domestic Product, Chained Dollars" (U.S. Bureau of Economic Analysis, 2025a). Given that the original series starts in 1929, I use the average annual growth rate for the period 1929–1939 to extrapolate backwards to year 1900.

Capital stock. There are three types of capital in the model: structures and equipment, and intellectual property products (IPP) capital, which is my main proxy for the stock of ideas in the economy. The three series are extracted from the Fixed Assets Accounts Tables from the BEA. In all the analysis, I exclude residential assets.[28] For each of the three series, I multiply the chain-type quantity indexes reported in Table 1.2 "Chain-Type Quantity Indexes for Net Stock of Fixed Assets and Consumer Durable Goods" (U.S. Bureau of Economic Analysis, 2025d), which equal 100 in the base year 2017, with the current cost stocks of 2017 obtained from Table 1.1 "Current Cost Net Stock of Fixed Assets" (U.S. Bureau of Economic Analysis, 2025c). In all cases, I take the aggregation of private and government values, reported in rows 18–20 of each of the tables. The resulting series

[28] Rognlie (2015) shows that housing is a major responsible for the documented decreasing in the labor share. The comment by Robert Solow published with the article suggests that an implication of Rognlie's results is that "for estimating an economy-wide elasticity of substitution, it would be better to eliminate the housing stock and associated land on the capital-input side and the rents on the output side." Excluding the part of GDP associated to housing for such a long time series is more complicated: it only appears disaggregated in the GDP measures starting in year 2002. However, not excluding this information from the GDP accounts is not particularly problematic in the context of my paper because the adjustments described below net out such aggregate discrepancies in the predicted level of wages by design.

are expressed in chained U.S. dollars of 2017. Given that the original series start in 1925, I extrapolate them backwards to 1900 using average annual growth rates.

Capital depreciation rates. The capital depreciation rates are computed combining the Fixed Assets Account Table 1.1 described above with the Table 1.3 "Current-Cost Depreciation of Fixed Assets and Consumer Durable Goods" U.S. Bureau of Economic Analysis (2025e). In particular, annual depreciation rates are obtained dividing the depreciation series (Table 1.3) for each type of capital by the current cost stock of capital (Table 1.1). Given that the original series start in 1925, I impute the 1925–1929 average to the period 1900–1924.

Relative prices of equipment. The relative price of equipment is obtained dividing capital stocks of structures and equipment in current cost (Fixed Assets Table 1.1, (U.S. Bureau of Economic Analysis, 2025c)), normalized to 100 in 2017, by their counterpats in real terms (Fixed Assets Table 1.2, (U.S. Bureau of Economic Analysis, 2025d)), and finally dividing the resulting equipment series by the one for structures. By construction, the resulting series equals 1 in 2017. Given that the original series start in 1925, I impute the 1925–1929 average to the period 1900–1928.

Cohort sizes. Cohort sizes are extracted from the Integrated Public Use Microdata Series (IPUMS) of the U.S. Census (Ruggles, Flood, Sobek, Backman, Cooper, Drew, Richards, Rodgers, Schroeder and Williams, 2025). In particular, the largest available sample from each decennial census for 1900–2000, and the American Community Survey (ACS) for 2001–2023. A person is classified as an immigrant if born abroad. Individuals born in Puerto Rico and other outlying areas are categorized as natives. I compute aggregates using person weights provided in the data.[29] Native and immigrant inter-census cohort sizes are estimated following different procedures. For natives, I distribute the decade's cohort size decrease across the years between censuses using annual mortality rates by age from Vital Statistics of the U.S.[30] For immigrants, I use a similar procedure to

[29] The 1970 Census includes the six available files, each representative of the whole U.S. Therefore, weights in that year are divided by six to represent the correct U.S. aggregates.

[30] Mortality rates are obtained from different sources: Table 6 in Linder and Grove (1943) for 1900–1940; Table 55 in Grove and Hetzel (1968) for 1941–1960; Tables 1.3 in National Center for Health Statistics (1974) for 1961–1970 and in National Center for Health Statistics (1985) for 1971–1980; Table 1.4 in National Center for Health Statistics (2002); and Tables 2 in Singh, Kochanek and MacDorman (1996), Anderson, Kochanek and Murphy (1997), Peters, Kochanek and Murphy (1998), Hoyert, Kochanek and Murphy (1999), Murphy (2000), and Hoyert, Arias, Smith, Murphy and Kochanek (2001) for each year between 1994 and 1999 respectively. The raw data is presented in pdfs, and were manually tabulated. Age is grouped in different categories. For each age, I apply the value of the corresponding aggregated age cell from the original data. In the period 1900–1940, reported rates refer to the subset of U.S. states that reported death rates.

distribute the net increase in the cohort sizes, but using the estimates of the entry age distribution described below instead of mortality rates.

Age at entry distribution. The distribution of entry age of immigrants is estimated using U.S. Census IPUMS. In order to reduce small sample noise, I average out the distributions for immigrants who arrived at $t-1$, $t-2$,..., $t-5$. Since the exact year of immigration is only available in 1900–1930 and 2000 Censuses, and in the ACS (2001–2023), intermediate years are linearly interpolated. Given that the distribution is stable over the years, I estimate a single distribution for each of the following intervals: 1900–1930, 1931–1940, 1941–1950, 1951–1960, 1961–1970, 1971–1980, 1981–1990 and 1991–2023. Finally, in order to obtain the joint distribution of age at entry and initial education, I estimate the entry age distribution conditional on education. Because of data limitations, I approximate it using the "relative" distribution by educational level: I compute the ratio of conditional and unconditional distributions from the Census 2000, and then I multiply this relative distribution by the time varying unconditional distribution of age at entry.

Regions of origin distribution. The share of immigrants from each region of origin is obtained from U.S. Census IPUMS 1900–2000 and ACS 2001–2023. I consider seven regions of origin for immigrants: Western Countries; Mexico; Central America and the Caribbean; South America; China and selected Southeast Asia; India; and Other Asia and Africa, which includes all the remaining immigrants from these two continents. Inter-census estimates of the distribution of immigrants from each region of origin are obtained by linear interpolation.

Ethnicity distribution. The share of natives in each ethnic group is obtained from the observed distributions in each U.S. Census IPUMS 1900-2000 and ACS 2001–2023 for individuals aged 16 to 25. I consider three ethnic groups for natives: Hispanic, Non-Hispanic Black, and Non-Hispanic Non-Black. Inter-census estimates of these shares are linearly interpolated.

Initial education distribution. The distribution of initial education is obtained from the U.S. Census IPUMS 1940–2000 and ACS 2001–2023. For natives, I take individuals aged 18 in each census, censoring the number of observed years of education at 10. I focus on 18-years-old individuals instead of 16 to allow for some individuals to temporarily drop out from school, something that is not allowed in the model, and to account for other sources of systematic reporting errors such as those derived from grade retention. The U.S. Census does not contain information on education prior to 1940, so information on earlier cohorts are obtained from the 1940 census, checking, respectively, individuals aged 28, 38, 48, and 58 respectively for years 1930, 1920, 1910, and 1900. For immigrants, I use data starting from 1970, since the 1940–1960 censuses do not include information on years since migration. For each 10-year cohort of arrival of immigrants I compute the observed education distribution, merging the data of different censuses. Because of the grouping of the information of years since migration, the distribution of education of immigrants that arrived before 1940 is assumed to be constant.

Fertility process. The fertility process consists of a three times three transition probability matrix, from 0, 1, or 2+ preschool children in period $t-1$ into 0, 1 or 2+ in t, conditional on age, college status, and demographic type. Given the model assumes a stationary distribution for the whole estimation period, I estimate two different processes: one for years before 1970, and and one for the years after 1970 (the transition from the first process to the second is unexpected in the model). The pre-1970 distribution is estimated using Census microdata for census years 1940–1970. The latter is estimated using the censuses of 1980–2000 and the ACS files for 2009–2011 and 2018–2023 to mimic one census point each (the weights of the ACS are divided by the number of ACS files pulled together to represent one year's U.S. population). Children are identified using the variables pernum, poploc, and momloc provided by IPUMS, as well as the household identifier. I infer transitions out of cross-sectional data based on the age of the child. In order to smooth probabilities across different ages, I estimate different logits on flexible polynomials of age for every number of children in $t-1$, observable type and college graduate status. Transitions from 0 to 2+ children and from 2+ to 0 are assumed to happen with probability zero.[31] Therefore, the probabilities are estimated using binary logits if the number of children in $t-1$ is 0 or 2+, and a multinomial logit in case it is 1. The logits are specified with up to fourth order polynomials in age. If they do not converge, the order of the polynomial is progressively reduced until convergence is achieved (the lowest specifications are with a quadratic polynomial). The order of the polynomial is also reduced if the largest terms are insignificant. In a few cases, when logits are to be run with less than 100 observations, the probability values are imputed as follows: if the number of children in $t-1$ was 2+, then it reduces to 1 with probability 1/6 and stays at 2+ with probability 5/6; if the number of children at $t-1$ was 1, it reduces to 0 with probability 1/6, it increases to 2+ with a probability that equals

[31] In practice, when I observe in the data an increase from 0 to 2+, I treat it as if this increase was from 0 to 1, and when a decrease is observed from 2+ to 0, the decrease is treated as if it was observed from 1 child to 0.

the probability that individuals in the same cell have of going from 0 to 1, and it stays at 1 with the remaining probability; if the number of children was 0, it stays at 0 with probability 1. After age 40, the probability of increasing the number of children is not allowed to grow, and, additionally, after age 45 the probability of reducing the number of children is not allowed to decrease.

Total wage bills by occupation. Total wage bills by occupation are constructed by combining CPS micro data from IPUMS (for the period 1967 to 2023) with aggregate BEA wage totals. In the CPS, I essentially follow the same approach to assign individuals to working in each occupation following the same approach as I describe below for the micro data. I also determine individual wages as described below. However, to compute annual hours for the years prior to 1975, I need to slightly adjust the approach, given data availability. In particular, weeks worked are only available intervalled, and usual hours per week worked last year are not available. For this reason, I use the average weeks for each interval based on the post 1975 data to impute weeks worked for each interval. Furthermore, I use hours worked last week instead of usual hours worked. Then, I aggregate wages at the year-occupation level to obtain the percentage occupational composition of earnings from the CPS. Finally, I implement the wage adjustments described below. Specifically, I multiply these shares to by the BEA aggregate total compensation of employees from NIPA Table 2.1 "Personal Income and Its Disposition" (U.S. Bureau of Economic Analysis, 2025b), and I correct by the third wage adjustment described below. This procedure preserves the CPS allocation of labor compensation across occupations while ensuring consistency with the aggregate wage bill used elsewhere in the model.

Wage adjustments. To avoid biases in parameter estimates, I make three adjustments to wages and/or aggregate skill units. On the one hand, there are two reasons why total labor compensation obtained from the aggregation of my CPS wage measurements may differ from the observed wages and salaries from the national accounts. First, given the discrete mutually-exclusive choices of the model and the annual frequency, individuals may be assigned to work full-time in an occupation all year when they worked part time or only during a fraction of the year. Likewise, some individuals assigned to school or home may have worked for part of the year. Second, there are forms of labor compensation that are not wages (e.g. some types of bonuses and in-kind payments). These two discrepancies are corrected by adjusting aggregate total wage compensation appropriately. To correct for the first, I adjust my predicted aggregate skill units multiplying them by the ratio between the total Wages and Salaries from NIPA Table 2.1 by the predicted aggregate wages from the CPS. To correct for the second, I multiply my skill prices by the ratio of BEA Compensation of Employees over Wages and Salaries, also from NIPA Table 2.1. On the other hand, the total labor share, which I use to determine the returns to assets, as discussed below, is obtained dividing BEA Compensation of Employees and Personal Income from NIPA Table 2.1. However, output is measured by GDP. Hence, I adjust the total wage bill by the ratio to GDP over Personal Income. For 1929–1967, for which CPS data are not available, I take the first adjustment as its value in 1967 and I let the other three move as in the data. When I do that, I deflate personal income using the ratio of disposable personal income in nominal and in real chained 2017 dollars corrected by the difference between this index and the IPUMS CPI index for 1967 to make the series consistent over time. For 1900–1928, I use the value of 1929.

Returns to assets. The implied returns to assets are obtained to be consistent with the model structure. I first measure the capital share as one minus compensation of employees divided by personal income, both taken from NIPA Table 2.1. I then back out the common return on assets from Equation (47). In line with the last wage adjustment described above, $\Gamma_{Kt}Y_t$ is obtained as the product of the capital share (one minus the labor share) and GDP (not personal income).

A2. Microdata

Labor supply parameters are estimated using micro data from the March Supplements of the Current Population Survey (CPS). The CPS data are obtained from IPUMS (Flood, King, Rodgers, Ruggles, Warren, Backman, Chen, Cooper, Richards, Schouweiler and Westberry, 2024). I selected the longitudinal Annual Social Economic Supplement files, linked by IPUMS across consecutive years. I extract those data from 1994–1995 to 2023–2024 (representing 1993–2023), which are the available years with information on nativity status. The second year corresponds to period t, and the first year is $t-1$. The variables referring to the latter are only used to clean the survey linkage, and to construct the $t-1$ choice d_{a-1}. To clean the linkage, I sequentially drop the observations that satisfy the following criteria: age difference across surveys is more than two years (40,843 observations); the reported sex differs across surveys (2,956 observations); immigrant status differs across surveys (1,228 observations); race differs across surveys (2,756 observations); region of origin differs across surveys (609 observations); year of immigration is unknown (24), and occupation is missing for a work candidate (3,268). Immigrant status, race, and region of origin are defined as follows.

Age. In line with the timing of the model, I keep individuals aged 16 to 65. To accurately fit with the model's timing, I make some small age adjustments to the reported age as a result of the education adjustments described below.

Demographic types. Individuals are classified in 20 types depending on sex, nativity, and race(for natives) or origin (for immigrants). Natives include individuals born in Puerto Rico and other U.S territories, and immigrants are individuals born elsewhere. The regions of origin for immigrants are defined above when describing their aggregate distributions. Natives are classified into nativity groups based on the race and Hispanic variables. The classification is done as follows. First, individuals are classified as Hispanic if they have a non-zero value in the hispan variable in the CPS (those who report "Do not know" or "N/A (and no response 1985-87" are not included as Hispanic). For the remaining individuals, those indicating black as their single race, as one of their multiple races, or as their ethnicity are classified as non-Hispanic black. Finally, the remaining individuals are classified as non-Hispanic non-black. The variable sex is used to further classify individuals into males and females.

Educational level. The main education variable is defined in years. In the CPS, school is provided in categories. For each category I assign the following years: 0 for "None or preschool"; 2.5 for "Grades 1, 2, 3, or 4"; 5.5 for "Grades 5 or 6"; 7.5 for "Grades 7 or 8"; 9 for "Grade 9"; 10 for "Grade 10"; 11 for "Grade 11" and for "12th grade no diploma"; 12 for "High school diploma or equivalent"; 14 for "Some college but not degree", for "Associate's degree, occupational/vocational", and for "Associate's degree, academic program"; 16 for "Bachelor's degree"; 17 for "Master's degree"; 20 for "Professional degree"; and 21 for "Doctorate degree". I then adjust age, education, and enrollment status to mimic the fact that in the model individuals who quit school are not allowed to return. The adjustments are as follows. If an individual reports to be enrolled in school in either $t-1$ or t, her education level is between 10 and 12 years, and her age is above education level plus six but below education level plus nine, I adjust age to education level plus six and the previous choice is defined to be school. If the individual reports to be enrolled in school in either $t-1$ or t, her education level is 7.5 or 9, and her age is respectively 16 or 16/17, I adjust age to be 16, education level to be 10, and previous choice to be school. If an individual reports the same education level in both $t-1$ and t, that education level is respected. In that case, if age is below what the education level would predict, I adjust as follows. If the individual reports to be enrolled in education last year or this year, I modify the age to reflect the minimum age that would allow for that education level and I ensure that the previous year decision was schooling. If the education level was referring to a range of years (e.g., some college), I additionally adjust exact years of education within the range to fit accordingly. If the gap in education is one education level and the individual reports to be enrolled in education last year or this year, I make analogous adjustments. For education levels in intervals, education in this case is assumed to be in the first year of the interval. For the remaining individuals that declare to be in school in either $t-1$ or t, I make the following choices. If education level is below 9 years or above 12 years I move their previous choice to not school and I keep their education and age. If their education level is 9 and age is below 18, the education level is increased to 10, and previous choice is considered to be school. If education is between 10 and 12 and their age is 20 or below, I keep their education level, I adjust the age as above, and I keep the previous year decision as school. For the remaining individuals, the previous choice is considered to be other than education and, if age is below years of education plus seven, it is increased to that level. Previous choice was set to non-school for all individuals of final ages different from education plus six or if the education level is 21 years.

Preschool children. Individuals are allowed to have 0,1, or 2+ preschool children (zero to five years old). This variable has been created by IPUMS: nchlt5.

Age at immigration. This variable is defined, for immigrants, as current age minus the difference between current year and year of immigration minus 16. When the resulting variable is smaller than 0 (i.e., entry before age 16), it is normalized to 0. In both datasets, some of the reported values correspond to (short) intervals. I keep the following values for intervals: 1949 for before 1950; 1951 for before 1952; 1954 fir before 1955; 1959 for before 1960; the center of the interval for periods of an odd number of years; and the first year above the center of the interval for periods of an even number of years.

Choices. Individuals are assigned to one of the five mutually exclusive year round alternatives: blue collar, white collar and white collar-STEM work, attend school, or stay at home. For each individual I need to compute current and lagged choices. The procedure to assign individuals follows a hierarchical rule. First, I assign individuals to education. Previous year education status is cleaned as described above when introducing the education level. Current choice is assigned to education if, on top of previous year being education, the individual reports to be enrolled in education this year. Individuals not assigned to education, are assigned to work or home following the following criteria. An individual is assigned to work in the preceding year if she worked at least 40 weeks (or at least 20 if last year's choice was school) and at least 20 (usual) hours per week. Individuals assigned to work are then classified into occupations based on their reported main occupation. To classify them into occupations, I first assign to STEM all individuals with a college degree that work in one of the occupations that the Bureau of Labor Statistics lists as STEM in any of its categories.[32] For the remaining workers, blue collar occupations include service workers, agricultural workers, construction workers, operatives, craftmen, and military, and white collar include non-STEM managers, professionals, scientists, technicians, and health workers, artists, sports and sales workers, and clerks.

Wages. Hourly wage is computed for individuals that are assigned to any of the work alternatives according to the previous definition. Workers are assumed to earn their wage entirely in the occupation they are assigned to. Earnings include wage and salary income, non-farm business income, and farm income. Wages are deflated to 2017 US dollars using the consumer price indices provided by IPUMS with the CPS data. Wages are divided by total hours worked (obtained as the product of usual weeks worked last year and usual hours worked last year), and multiplied by 2080, which is the corresponding amount for year-round full-time work. A few extreme observations earning less than 1 dollar or more than 200 dollars per hour are set to missing.

[32] `https://www.bls.gov/soc/Attachment_B_STEM.pdf`, accessed on September 30th, 2022.

Appendix B: Simulation of Idiosyncratic Shocks

This appendix describes how to simulate idiosyncratic errors ε_a from the distribution described in (13) along with productivity shocks η_a and taste shocks ϵ_{ja} for $j \in \{B, W, T, S, H\}$. Since $\varepsilon_{Sa} = \epsilon_{Sa}$ and $\varepsilon_{Ha} = \epsilon_{Ha}$ are independent of the other shocks and of η_a, they are trivially obtained transforming uniform draws by the quantile function of the Type-I extreme value distribution. That is, let q_S and q_H denote two independent draws from a standard uniform, $\mathbb{U}(0,1)$, then the corresponding draws of ε_S and ε_H are obtained as:

$$\varepsilon_j = -\ln(-\ln q_j), \quad \text{for } j \in \{S, H\}. \tag{B1}$$

To simulate draws for ε_j for $j \in \{B, W, H\}$, I first derive the marginal distribution of ε_B, the conditional distribution of ε_W given ε_B, and the one of ε_T conditional on the other two. Let $f_{\varepsilon_B\varepsilon_W\varepsilon_T}(\varepsilon_B, \varepsilon_W, \varepsilon_T)$ denote the joint probability density function of ε_B, ε_W, and ε_T, and let $F_{\varepsilon_B\varepsilon_W\varepsilon_T}(\varepsilon_B, \varepsilon_W, \varepsilon_T)$ denote their cumulative distribution function. Equation (13) implies that:

$$\begin{aligned} F_{\varepsilon_B\varepsilon_W\varepsilon_T}(\varepsilon_B, \varepsilon_W, \varepsilon_T) &= \exp\left\{-\left(e^{-\varepsilon_B/\varrho} + e^{-\varepsilon_W/\varrho} + e^{-\varepsilon_T/\varrho}\right)^{\varrho}\right\} \\ &\equiv \exp\left\{-\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho}\right\}, \end{aligned} \tag{B2}$$

and:

$$\begin{aligned} &f_{\varepsilon_B\varepsilon_W\varepsilon_T}(\varepsilon_B, \varepsilon_W, \varepsilon_T) \\ &\quad= \frac{\partial^3}{\partial\varepsilon_B\partial\varepsilon_W\partial\varepsilon_T} \exp\left\{-\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho}\right\} \\ &\quad= \frac{\partial^2}{\partial\varepsilon_W\partial\varepsilon_T} \exp\left\{-\left(\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho} + \frac{\varepsilon_B}{\varrho}\right)\right\} \mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{(\varrho-1)} \\ &\quad= \frac{\partial}{\partial\varepsilon_T}\left[\begin{array}{l} \exp\left\{-\left(\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho} + \frac{\varepsilon_B+\varepsilon_W}{\varrho}\right)\right\} \mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{2(\varrho-1)} \\ \times\left(1 - \frac{\varrho-1}{\varrho}\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{-\varrho}\right) \end{array}\right] \\ &\quad= \exp\left\{-\left(\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho} + \frac{\varepsilon_B+\varepsilon_W+\varepsilon_T}{\varrho}\right)\right\} \mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{3(\varrho-1)} \\ &\qquad \times\left(1 - \frac{\varrho-1}{\varrho}\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{-\varrho}\left[3 - \frac{\varrho-2}{\varrho}\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{-\varrho}\right]\right) \end{aligned} \tag{B3}$$

Given this expression, the marginal density function for ε_B is given by:

$$\begin{aligned} f_{\varepsilon_B}(\varepsilon_B) &\equiv \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{\varepsilon_B\varepsilon_W\varepsilon_T}(\varepsilon_B, \varepsilon_W, \varepsilon_T)d\varepsilon_W d\varepsilon_T \\ &= \exp\left\{-\left(\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho} + \frac{\varepsilon_B}{\varrho}\right)\right\} \mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{(\varrho-1)}\Big]_{-\infty}^{+\infty}\Big]_{-\infty}^{+\infty} \\ &= \exp\{-(e^{-\varepsilon_B} + \varepsilon_B)\}, \end{aligned} \tag{B4}$$

which is the probability density function of a Type-I Extreme Value distribution. Thus, ε_B is drawn using the quantile function in (B1). Having drawn it, we now need to draw from the conditional distribution of ε_W given ε_B. To compute it, we first need to derive the joint distribution of ε_B and ε_W, which is given by:

$$\begin{aligned}
f_{\varepsilon_B \varepsilon_W}(\varepsilon_B, \varepsilon_W) &\equiv \int_{-\infty}^{\infty} f_{\varepsilon_B \varepsilon_W \varepsilon_T}(\varepsilon_B, \varepsilon_W, \varepsilon_T) d\varepsilon_T \\
&= \left[\begin{array}{l} \exp\left\{ - \left(\mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{\varrho} + \frac{\varepsilon_B + \varepsilon_W}{\varrho} \right) \right\} \mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{2(\varrho-1)} \\ \times \left(1 - \frac{\varrho - 1}{\varrho} \mathcal{Q}(\varepsilon_B, \varepsilon_W, \varepsilon_T)^{-\varrho} \right) \end{array} \right]_{-\infty}^{+\infty} \\
&= \exp\left\{ - \left[\mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho} + \frac{\varepsilon_B + \varepsilon_W}{\varrho} \right] \right\} \mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho - 2} \\
&\quad \times \left(\mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho} + \frac{1 - \varrho}{\varrho} \right), \qquad \text{(B5)}
\end{aligned}$$

where $\mathcal{Q}_2(\varepsilon_B, \varepsilon_W) \equiv \exp(-\varepsilon_B/\varrho) + \exp(-\varepsilon_W/\varrho)$. Thus, the conditional density is:

$$\begin{aligned}
f_{\varepsilon_W | \varepsilon_B}(\varepsilon_W | \varepsilon_B) &\equiv \frac{f_{\varepsilon_B \varepsilon_W}(\varepsilon_B, \varepsilon_W)}{f_{\varepsilon_B}(\varepsilon_B)} \\
&= \exp\left\{ e^{-\varepsilon_B} + \varepsilon_B - \left[\mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho} + \frac{\varepsilon_B + \varepsilon_W}{\varrho} \right] \right\} \\
&\quad \times \mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho - 2} \left(\mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho} + \frac{1 - \varrho}{\varrho} \right). \qquad \text{(B6)}
\end{aligned}$$

The conditional cumulative function is obtained integrating the above expression with respect to ε_W, which yields:

$$\begin{aligned}
F_{\varepsilon_W | \varepsilon_B}(\varepsilon_W | \varepsilon_B) &\equiv \int_{-\infty}^{\varepsilon_W} f_{\varepsilon_W | \varepsilon_B}(\varepsilon | \varepsilon_B) d\varepsilon \qquad \text{(B7)} \\
&= \exp\left\{ e^{-\varepsilon_B} + \varepsilon_B - \left[\mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho} + \frac{\varepsilon_B}{\varrho} \right] \right\} \mathcal{Q}_2(\varepsilon_B, \varepsilon_W)^{\varrho - 1}.
\end{aligned}$$

Inverting this function with respect to ε_W to obtain the conditional quantile function, and evaluating it on a uniform random draw q_W, we simulate ε_W as:

$$\varepsilon_W = -\varrho \ln \left\{ -e^{-\varepsilon_B/\varrho} + \left[\frac{1 - \varrho}{\varrho} \mathbb{W} \left(\frac{\varrho}{1 - \varrho} \exp\left\{ \tfrac{\varrho}{1 - \varrho} e^{-\varepsilon_B} - \varepsilon_B \right\} q_W^{\varrho/(\varrho - 1)} \right) \right]^{1/\varrho} \right\}, \qquad \text{(B8)}$$

where $\mathbb{W}(\cdot)$ is the Lambert-W function or product logarithm function, which is defined as the inverse of the function $f(x) = xe^x$ (or equivalently as $\mathbb{W}(y)e^{\mathbb{W}(y)} = y$).

Finally, the conditional distribution of ε_T given ε_B and ε_W is given by:

$$\begin{aligned}
& f_{\varepsilon_T|\varepsilon_B\varepsilon_W}(\varepsilon_T|\varepsilon_B,\varepsilon_W) \\
& \equiv \frac{f_{\varepsilon_B\varepsilon_W\varepsilon_T}(\varepsilon_B,\varepsilon_W,\varepsilon_T)}{f_{\varepsilon_B\varepsilon_W}(\varepsilon_B,\varepsilon_W)} \\
& \quad = \exp\left\{\mathcal{Q}_2(\varepsilon_B,\varepsilon_W)^\varrho - \mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^\varrho - \frac{\varepsilon_T}{\varrho}\right\}\mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^{3(\varrho-1)} \\
& \qquad \times \left(1 - \frac{\varrho-1}{\varrho}\mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^{-\varrho}\left[3 - \frac{\varrho-2}{\varrho}\mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^{-\varrho}\right]\right) \\
& \qquad \times \mathcal{Q}_2(\varepsilon_B,\varepsilon_W)^{2(\varrho-1)}\left(1 - \frac{\varrho-1}{\varrho}\mathcal{Q}_2(\varepsilon_B,\varepsilon_W)^{-\varrho}\right)^{-1}. \qquad \text{(B9)}
\end{aligned}$$

Integrating this expression with respect to ε_T we obtain:

$$\begin{aligned}
F_{\varepsilon_T|\varepsilon_B,\varepsilon_W}(\varepsilon_T|\varepsilon_B,\varepsilon_W) \equiv & \int_{-\infty}^{\varepsilon_T} f_{\varepsilon_T|\varepsilon_B\varepsilon_W}(\varepsilon|\varepsilon_B,\varepsilon_B)d\varepsilon \\
& = \exp\{\mathcal{Q}_2(\varepsilon_B,\varepsilon_W)^\varrho - \mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^\varrho\} \qquad \text{(B10)} \\
& \quad \times \frac{(1-\varrho)\mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^{2\varrho-3} + \varrho\mathcal{Q}(\varepsilon_B,\varepsilon_W,\varepsilon_T)^{3(\varrho-1)}}{(1-\varrho)\mathcal{Q}_2(\varepsilon_B,\varepsilon_W)^{2\varrho-3} + \varrho\mathcal{Q}_2(\varepsilon_B,\varepsilon_W)^{3(\varrho-1)}}.
\end{aligned}$$

Finally, the conditional quantile function, which does not have a closed form solution, is used to transform uniform draws into draws from $F_{\varepsilon_T|\varepsilon_B\varepsilon_W}(\varepsilon_T|\varepsilon_B,\varepsilon_W)$.

All the steps so far show how to simulate ε_B, ε_W, and ε_B from its joint distribution. A prior step is needed to complete the simulation. In particular, we need to independently draw ϵ_B from its unknown distribution and $\sigma_{B\ell}\Lambda_0\eta_a$ from a normal distribution, and then obtain ε_B as the sum of the two (which by construction will make it a Type-I extreme value). Having done that, we should then proceed simulating ε_{Wa} and ε_{Ta} as described in this Appendix. And finally, the remaining taste shocks are obtained as functions of these draws as $\epsilon_{Wa} = \varepsilon_{Wa} - \sigma_{W\ell}\Lambda_0\eta_a$ and $\epsilon_{Ta} = \varepsilon_{Ta} - \sigma_{T\ell}\Lambda_0\eta_a$.

The remaining unknown is the distribution of the taste shock ϵ_{Ba}, $f_{\epsilon_B}(\epsilon_B)$. This distribution is obtained as the deconvolution of a Type-I Extreme Value and a normal distribution with zero mean and variance $\sigma_{B\ell}\Lambda_0$. This deconvolution is derived using the characteristic functions of the two distributions which are, respectively, $\mathbb{G}(1-it)$ and $e^{-\frac{t^2}{2}\sigma_{B\ell}^2\Lambda_0}$, where $\mathbb{G}(\cdot)$ denotes the complete gamma function, $\mathbb{G}(x) \equiv \int_0^\infty z^{x-z}e^{-z}dz$, and i denotes the imaginary unit. In particular:

$$f_{\epsilon_B}(\epsilon_B) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathbb{G}(1-it)e^{-\frac{t^2}{2}\sigma_{B\ell}^2\Lambda_0^2 - it\epsilon_B}dt, \qquad \text{(B11)}$$

which does not have a closed form. The cumulative density function and the quantile function are then computed numerically.